Part III: Geometric Representation Theory

Yutong Chen
(based on lectures by Prof. Ian Grojnowski)

1 Introduction

Exercise. Prove the following statements:

1. All choices of Hermitian inner product on $\mathbb{C}^2$ are conjugate. So $\mathrm{SU}_2 \to \mathrm{SL}_2$ is defined up to conjugation.
2. $\mathrm{SU}_2 = S^3$
3. $\mathrm{SL}_2(\mathbb{C}) \cong T\mathrm{SU}_2 = TS^3$ diffeomorphism. In particular SU_2 is a maximal compact subgroup of $\mathrm{SL}_2(\mathbb{C})$, and all maximal compact subgroups of $\mathrm{SL}_2(\mathbb{C})$ arise in this way.
4. $\mathrm{GL}_n(\mathbb{C}) = U_n \times \exp(\mathfrak{p})$, for $\mathfrak{p} = \{A \in M_n(\mathbb{C}) : A = \overline{A^t}\}$. This is called the polar decomposition

Remark 1.1. *We will see that every simply-connected* ***simple algebraic group*** *(a connected, semisimple linear algebraic group such that every proper connected normal subgroup is is trivial) has a unique conjugacy class of maximal compact subgroups.*

Theorem 1.2. *Let k algebraically closed field of characteristic 0.*

1. *All irreducible algebraic representations of $\mathrm{SL}_2(k)$ are finite dimensional.*
2. *Every finite dimensional representation is completely reducible.*
3. *The irreducible representation of are $S^n(k^2)$ for $n \geq 0$. As a representation of $\mathrm{SL}_2 \times \mathrm{SL}_2$, $k[\mathrm{SL}_2] = \bigoplus_{n\geq 0} S^n k^2 \otimes (S^n k^2)^*$*

Theorem 1.3. *Let $k = \mathbb{C}$.*

1. *Every continuous irreducible representation of SU_2 is finite dimensional.*
2. *Every finite dimensional representation of SU_2 is completely reducible.*
3. *If V is an irreducible representation fo $\mathrm{SL}_2(\mathbb{C})$, then it remains to be irreducible as a SU_2 representation.*
4. *All irreducible representations of SU_2 arise in this way.*

Theorem 1.4. *Let $k = \mathbb{C}$.*

1. *Every finite dimensional representation of $\mathfrak{sl}_2(\mathbb{C})$ is completely reducible.*
2. *For (V, ρ) irreducible representation of SL_2, $d\rho$ is an irreducible representation of $\mathfrak{sl}_2$*
3. *Every finite dimensional representation of $\mathfrak{sl}_2$ arises in this way.*

These representations come from the equivariant actions of G on the line bundles of the associate flag variety, and hence the group G acts the cohomology of the line bundles. But first of all we need to understand the line bundles on the projective spaces and their cohomology.

On $\mathbb{P}(V)$, we have line bundles $\mathcal{O}(n)$. We have the interpretation

$$\mathcal{O}(n)_l = \{f : l \to k | f(\lambda v) = \lambda^n f(v), \forall v \in l, \lambda \in k\} = (l^*)^{\otimes n} = (l^{\otimes n})^*$$

We wish to let GL_V acts equivariantly on $\mathcal{O}(n) : g \cdot \mathcal{O}(n)_l \cong \mathcal{O}(n)_{gl}$. So GL_V acts on $\Gamma(\mathbb{P}V, \mathcal{O}(n))$ and even on $H^i(\mathbb{P}, \mathcal{O}(n))$. But we have the following explicit computations:

$$H^0(\mathbb{P}V, \mathcal{O}(n)) = \begin{cases} S^n V^*, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

$$H^{\dim V - 1}(\mathbb{P}V, \mathcal{O}(n)) = \begin{cases} 0, & n \geq -d+1 \\ S^{-n-d}V \otimes \det V, & n \leq -d \end{cases}$$

We observe that the irreducible representations of SL_2 are parametrized by global sections of semiample line bundles on $\mathbb{P}^1$ (note higher cohomology vanishes in this case). However, if $\dim V > 2$, we still know $S^n V^*$ are irreducible, but these don't exhaust all the irreducible representations.

The cohomology of quasi-coherent sheaves on $\mathbb{P}^1$ can be explicitly described as following. Let $\mathbb{P}^1 = \mathbb{P}V$, where $V^* = \langle x, y \rangle$. Set $U_0 = \{y \neq 0\}, U_\infty = \{x \neq 0\}, U_{0,\infty} = U_0 \cap U_\infty$. We know $\mathbb{P}_1 = U_0 \cup U_\infty$. Let $\mathcal{M} \in \mathrm{QCoh}(\mathbb{P}^1)$, write $M_i = \Gamma(U_i, \mathcal{M})$. We have a restriction map $M_0 \oplus M_\infty \xrightarrow{\alpha} M_{0,\infty}$. Then $H^0(\mathbb{P}^1, \mathcal{M}) = \ker \alpha; H^1(\mathbb{P}^1, \mathcal{M}) = \mathrm{coker}\ \alpha$.

Example 1.5. *We look at the sheaf $\mathcal{O}(-n[0])$ of functions on $\mathbb{P}^1$ vanishing at 0 in order $n > 0$. The quasi-coherent sheaf is given by $z^n k[z] \oplus k[z^{-1}] \xrightarrow{\alpha} k[z, z^{-1}]$. One sees that $\Gamma(\mathbb{P}^1, \mathcal{O}(-n[0])) = 0$, and $H^1(\mathbb{P}^1, \mathcal{O}(-n[0]))$ has a basis $\{z, z^2, \cdots, z^{n-1}\}$ for $n \geq 2$. Note that $\mathcal{O}(-n[0]) \cong \mathcal{O}(-n)$ abstractly, but it does not admit an equivariant SL_2 action.*

Example 1.6. *We consider the line bundle $\mathcal{O}(n)$. We know $\mathcal{O}(n)_l = k \cdot y^n$, for $l = \{x = 0\} \in \mathbb{P}^1$. Set $z = \frac{x}{y}, U_0 = \{y \neq 0\}, U_1 = \{x \neq 0\}$. We have: $\mathcal{O}(n)|_{U_0} = y^n k[z], \mathcal{O}(n)|_{U_1} = x^n k[z^{-1}]$. To compute cohomology, we have the following short exact sequence:*

$$0 \to H^0(\mathbb{P}^1, \mathcal{O}(n)) \to (y^n k[z] \oplus x^n k[z^{-1}] \to x^n k[z, z^{-1}]) \to H^1(\mathbb{P}^1, \mathcal{O}(n)) \to 0$$

1. *If $n \geq 0$, $H^0(\mathbb{P}^1, \mathcal{O}(n))$ has a basis $\{v_+ = (y^n z^n, x^n), \cdots, v_- = (y^n, x^n z^{-n})\}$;*
2. *If $n \leq -2$, $H^1(\mathbb{P}^1, \mathcal{O}(n))$ has a basis $\frac{1}{xy}(v_- = x^{n+2}, \cdots, v_+ = y^{n+2})$.*

There is a natural identification of S^nV^* with $H^0(\mathbb{P}^1, \mathcal{O}(n))$.

Now consider the $G = \mathrm{SL}_2$-action on $\mathbb{P}^1$. Let B be the stablizer of point infinity $\begin{pmatrix}1\\0\end{pmatrix} \implies B = \begin{pmatrix}* & *\\0 & *\end{pmatrix}$. We note that $\mathbb{P}^1 = G/B$, is a homogeneous space for G. There is a natural action of B on $\mathcal{O}(n)_\infty = kx^n$:

$$\begin{pmatrix}a & b\\0 & a^{-1}\end{pmatrix} x^n = a^{-n}x^n$$

So we can view $\mathcal{O}(n)_\infty$ as a representation of B, which factors through the representation λ_{-n} of the maximal torus T. We shall see that the representation the global section functor is indeed the "induction" on this representation on B to G.

Exercise. Let $G = \mathrm{GL}_V, d = \dim V \geq 2$, then G acts on $\mathbb{P}V$.

$$\mathrm{stab}_G \begin{pmatrix}1\\0\\0\\0\end{pmatrix} = \begin{pmatrix}* & * & * & *\\0 & * & * & *\\0 & * & * & *\\0 & * & * & *\end{pmatrix} = P_d$$

So $\mathbb{P}^{d-1} = \mathrm{GL}_d/P_d$ as a set (in fact as a variety). The problem here is that P_d is not solvable, so has irreducible representations which are not of dimension 1.

Theorem 1.7 (Borel-Weil-Bott)**.** *Let G be a simple algebraic group, B its Borel subgroup (i.e. maximal solvable closed connected subgroup). Let $\mathcal{B} = G/B$ be the associated flag variety.*

1. *$\mathcal{B}$ is projective.*
2. *every irreducible representation of G arises uniquely as $\Gamma(\mathcal{B}, \mathcal{L})$, for $\mathcal{L}$ a G-equivariant line bundle on $\mathcal{B}$.*

Exercise. Let $G = \mathrm{SL}_V = \mathrm{SL}_d, d = \dim V$. Let $B = \{\text{upper triangular matrices}\}$, $\mathcal{B}$ the associated flag variety of G.

1. G acts transitively on $\mathcal{B}$, with the stablizer of $\{\langle e_1\rangle, \cdots, \langle e_1, \cdots, e_d\rangle\}$ is B. This shows $\mathcal{B} \cong G/B$.
2. $\mathcal{B}$ is projective. (Hint: embed $\mathcal{B}$ into Grassmannians, which are projective from Plücker embedding)
3. Pic $\mathcal{B} = \mathbb{Z}^{d-1}$, and every $\mathcal{L}$ is naturally G-equivariant.
4. Show the ample cones are bijective to the dominant integral weights.
5. Do the same for $\mathrm{Sp}(V)$, where $\dim V = 2d, \mathcal{B} = \{F_1 \leq F_2 \cdots \leq F_d | \dim F_i = i, F_i \subseteq F_i^\perp\}$.

Example 1.8. *We carefully walk through the case for $G = \mathrm{SL}_3$. $\mathcal{B} = \{(F_1 < F_2) | \dim(F_i) = i\} = \{(v, v^*) \in \mathbb{P}V \times \mathbb{P}V^* | \langle v^*, v\rangle = 0\}$. Hence there is a canonical map $i : \mathcal{B} \to \mathbb{P}^2 \times \mathbb{P}^2$.*

Lefschetz Hyperplane Theorem tells us that any line bundle on $\mathcal{B}$ is of form:

$$i^*(\mathcal{O}(a) \boxtimes \mathcal{O}(b)) \coloneqq \mathcal{L}_{a,b}$$

Let $\mathcal{I}_{\mathcal{B}}$ be the ideal of functions vanishing on $\mathcal{B}$ (the defining equations of $\mathcal{B}$):

$$\left\langle s = \sum_{i=0}^{2} x_i y_i \in \Gamma(\mathbb{P}^2 \times \mathbb{P}^2, \mathcal{O}(1) \boxtimes \mathcal{O}(1)) \right\rangle$$

Hence we have $\mathcal{O}_{\mathbb{P}^2 \times \mathbb{P}^2} = \mathcal{O} \xrightarrow{s} \mathcal{O}(1) \boxtimes \mathcal{O}(1)$. Dualizing, we get a short exact sequence:

$$0 \to \mathcal{O}(-1) \boxtimes \mathcal{O}(-1) \to \mathcal{O} \to i_* \mathcal{O}_{\mathcal{B}} \to 0$$

To compute the cohomology of $\mathcal{L}_{a,b}$, we push it forward to $\mathbb{P}^2 \times \mathbb{P}^2$: $i_* \mathcal{L}_{a,b}$. Tensoring the above exact sequence with $i_* \mathcal{L}_{a,b}$, we get:

$$0 \to \mathcal{O}(a-1) \boxtimes \mathcal{O}(b-1) \to \mathcal{O}(a) \boxtimes \mathcal{O}(b) \to i_* \mathcal{L}_{a,b} \to 0$$

We can then use this (and snake lemma) to compute cohomologies of $i_* \mathcal{L}_{a,b}$. In particular: if $a, b \geq 0$,then all higher cohomology vanishes and $\Gamma(\mathcal{B}, \mathcal{L}_{a,b}) = \operatorname{coker}(L_{a-1} \otimes L_{b-1} \to L_a \otimes L_b)$, where $L_a = S^a V^* = \Gamma(\mathbb{P}^2, \mathcal{O}(a))$, for $V = k^3$.

Exercise. Assume we're in the setup of SL_3.

1. Check this is correct by decomposing $L_a \otimes L_b$ as $\mathfrak{sl}_3$-representations.
2. Compute $\dim \Gamma(\mathcal{B}, \mathcal{L}_{a,b})$. Check this agrees with Weyl dimension formula.
3. Compute Higher cohomology of $H^i(\mathcal{B}, \mathcal{L}_{a,b})$ for arbitrary a, b.
4. (bonus) There is a fibration $\mathcal{B} \to \mathbb{P}^2$ with fibre $\mathbb{P}^1$. Use this to compute the global section of $\mathcal{L}_{a,b}$.

We now consider the Lie group picture. Let $\mathrm{SU}_V \subset \mathrm{SL}_V$ acts on $\mathbb{P}(V)$.

Exercise. 1. Check the above action is transitive, and stab $\begin{bmatrix} 1 \\ 0 \\ \dots \\ 0 \end{bmatrix} = P_d \cap \mathrm{SU}_V = U_{d-1}$.

So, $\mathrm{SU}_V / U_{d-1} \cong \mathrm{SL}_d / P_d = \mathbb{P}V$.

2. for $d = 2$, the right fibration is homotopic to the left one (the Hopf fibration)

$$\begin{array}{ccc} S^1 & \longrightarrow & \mathrm{SU}_2 \\ & & \downarrow \\ & & \mathrm{SU}_2/S_1 = S^2 \end{array} \qquad \begin{array}{ccc} B & \longrightarrow & \mathrm{SL}_2 \\ & & \downarrow \\ & & \mathbb{P}^1 \end{array}$$

3. Let SU_V acts on $\{v \in V | \langle v, v\rangle = 1\} = S^{2d-1}$. This gives a fibration:

$$\begin{array}{ccc} \mathrm{SU}_{d-1} & \longrightarrow & \mathrm{SU}_d \\ & & \downarrow \\ & & S^{2d-1} \end{array}$$

This realizes SU_d as "twisted product" of $S^3, S^5, \cdots, S^{2d-1}$.

Finally we remark on the Lie algebra picture. Let $G = \mathrm{SL}_2$ act on $\mathbb{P}^1$. This induces the Lie algebra $\mathfrak{g}$ acting by derivations on the vector field $\Gamma(\mathbb{P}^1, T\mathbb{P}^1)$.

Let $B = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix}, \mathfrak{b} = \text{Lie } B, \mathfrak{g} = \mathfrak{sl}_2$. We have the Verma module of highest weight λ: $M_\lambda = U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} \mathbb{C}_\lambda$. It has a basis: $\{F^k v_\lambda\}, Hv_\lambda = \lambda v_\lambda$.

If $\lambda = n \in \mathbb{N}$, we have:

$$M_{-n-2} \hookrightarrow M_n \twoheadrightarrow L_n$$

, where $L_n \cong S^n\mathbb{C}^2$ of highest weight n (one can see this by Weyl character formula).

The problem is that the choice of the Borel is non canonical. In fact a choice of Borel B is equivalent to a choice of a point in $\mathbb{P}^1$. This gives us a continuous 2 parameter family (highest weight, and the choice of Borel) of indeomposable representations of $\mathfrak{sl}_2$.

2 Affine Algebraic Groups

2.1 Definition and Basic Properties

Definition 2.1 (Algebraic Groups)**.** An **algebraic group** is an algebraic variety G together with morphism of varieties:

$$\mu : G \times G \to G, (x, y) \mapsto \mu(x, y)$$

$$i : G \to G, x \mapsto x^{-1}$$

satisfying the standard group axioms.

If the variety is in fact affine, we call the algebraic group a **linear algebraic group**.

A morphism between algebraic groups is a group homomorphisms that is also a morphism of algebraic varieties.

Example 2.2. *1. the multiplication group* $\mathbb{G}_m = \text{Spec } k[x, x^{-1}] = (k^*, \times)$.

2. the additive group $\mathbb{G}_a = \text{Spec } k[x] = (k, +)$.

3. Any finite group.

4. Non-example: elliptic curves.

Remark 2.3. *There are a few standard constructions:*

1. For two algebraic groups G, H*, then* $G \times H$ *has a natural algebraic group structure.*

2. *If H is a closed subgroup of G in terms of the Zariski topology, then there is a unique algebraic group structure on H making the inclusion $H \hookrightarrow G$ a morphism of algebraic groups.*

Let G be a linear algebraic group, and $A = k[G]$ be its coordinate ring. Then the group operations μ and i give rise to algebra homomorphisms $\Delta : A \to A \otimes A$ (the **comultiplication**) and $\iota : A \to A$ (the **antipode**). Moreover, the identity element of G is a morphism $\{\text{pt}\} \to G$, this corresponds to an algebra morphism $\varepsilon : A \to k$ (the **counit**), and let η be the inclusion map $k \to A$ (the **unit**). One can translate the group axioms into properties of the algebra $A = k[G]$ in an obvious way. We therefore get the data $(A = k[G], m, \Delta, \eta, \varepsilon, \iota)$, where m is the multiplication of k-algebra A. The axioms say that $A = k[G]$ is a **commutative Hopf algebra**.

Example 2.4. *Let $G = \mathbb{G}_a = \text{Spec } k[x]$. We carefully work out the Hopf algebra structure on $k[G] = k[x]$.*

1. *The multiplication $\mu : \text{Spec } k[x, y] \to \text{Spec } k[x]$ induces $\Delta : k[x] \to k[x, y], x \mapsto x \circ \mu$. Therefore $\Delta(x)(a, b) = x(a + b) = a + b$, which means $\Delta(x) = x + y$.*

2. *The inverse $i : \text{Spec } k[x] \to \text{Spec } k[x]$ induces $\iota : k[x] \to k[x], x \mapsto x \circ i$. So $\iota(x)(a) = x(-a) = -a, \implies \iota(x) = -x$.*

3. *The identiy $\text{Spec } k \to \text{Spec } k[x]$ induces $\varepsilon : k[x] \to k, x \mapsto (a \mapsto x(0) = 0) = 0$.*

This fully describes the Hopf algebra $k[x]$.

Example 2.5. *Let $G = \text{GL}_n(k)$. We show that G is indeed an algebraic group and describe the Hopf algebra structure of $k[G] = k[T_{ij}, D^{-1}]$, where $D = \det(T_{ij})$ is the determinant function.*

1. *$\Delta : k[G] \to k[G] \otimes k[G], T_{ij} \to T_{ij} \circ \mu$, where $T_{ij} \circ \mu(P, Q) = (PQ)_{ij} \implies \Delta T_{ij} = \sum_{h=1}^{n} T_{ih} \otimes T_{hj}$.*

2. *$\iota T_{ij} = \{G \to k, g \mapsto (g^{-1})_{ij}\}$*

3. *$\varepsilon : T_{ij} \mapsto \delta_{ij}$*

Let X be an affine variety, and G acts on X: $G \times X \xrightarrow{a} X$. We thus have an induced coaction $a^* : k[X] \to k[G] \otimes k[X]$. Note one can easily rephrase the action in terms of properties of $a^*, k[G], k[X]$. Moreover, note that since G acts on X, G acts on $k[X]$ naturally.

Definition 2.6. We call V a locally finite G-module if $V = \bigcup V_\alpha, G \cdot V_\alpha \subseteq V_\alpha$, where each V_α is finite dimensional over k.

Lemma 2.7 (Key Finiteness Lemma)**.** *Let G be an affine algebraic group, X affine variety. Then the G-action on $k[X]$ is locally finite*

Example 2.8. *Let $G = \mathbb{G}_a$, we study the G-action on $k[G]$. The comultiplication $\Delta : k[G] \to k[G] \otimes k[G], x \mapsto x \otimes 1 + 1 \otimes x$. We compute that $\Delta(x^n) = \sum \binom{n}{i} x^{n-i} \otimes x^i$, and by definition $\Delta(f)(x, y) = f(x + y)$. Now the action of G on $k[G]$ is given by $(g \cdot f)(h) = f(g^{-1}h)$, in our case $\lambda \circ f(x) = f(x - \lambda)$. But note that $J_{n+1} = \langle 1, x, \cdots, x^n \rangle$ is G-stable, and $k[G] = \bigcup_{n \geq 0} J_{n+1}$.*

Example 2.9. *Study the $G = \mathbb{G}_m$-action on $k[G]$. $\Delta(x) = x \otimes x, \Delta(f)(x,y) = f(xy)$. We note that kx^n is G-stable, and therefore $k[G] = \bigoplus kx^n$ is locally finite.*

Proof. Consider the coaction $a^* : k[X] \to k[G] \otimes k[X]$. Let $f \in k[X]$, and write $a^*(f) = \sum_{i=1}^r h_i \otimes \gamma_i$, for r minimal . We know that $(a^*f)(g,x) = f(gx) = \sum_{i=1}^r h_i(g)\gamma_i(x)$. In particular $f(x) = \sum_{i=1}^r h_i(1)\gamma_i(x)$. $(g^{-1}f)(x) = \sum_{i=1}^r h_i(g)\gamma_i(x)$, this shows $g^{-1}f \in \langle \gamma_1, \cdots, \gamma_r \rangle$ for any $g \in G$. We therefore only need to show that $\langle \gamma_1, \cdots, \gamma_r \rangle$ is G-stable.

It suffices to find $g_1, \cdots, g_r$ such that $h_i(g_j)$ has nonzero determinant. If so we can invert $\gamma_1, \cdots, \gamma_r \in \left\langle g_1^{-1}f, \cdots, g_r^{-1}f \right\rangle$. By minimality of r, we know $h_1, \cdots, h_r$ are linearly independent over k. We now perform induction on r. The case for $r = 1$ is clear.

We consider the matrix

$$\begin{pmatrix} h_1 & \cdots & h_r \\ h_1(g_2) & \cdots & h_r(g_2) \\ \cdots & \cdots & \cdots \\ h_1(g_r) & \cdots & h_r(g_r) \end{pmatrix} = \sum \alpha_i h_i, \alpha_i = \alpha_i(g_2, \cdots, g_r)$$

By induction we choose $g_2, \cdots, g_r$ such that $\alpha_1 \neq 0$. We can than choose g_1 appropriately. This proves for the case for $k[G]$ reduced. Otherwise we consider the algebraic group $k[G]/\text{nilradical}$, which is an affine reduced algebraic group. □

Corollary 2.10. *G is affine if and only if G is linear.*

Proof. If G is linear, G is obviously affine. We need to show the converse. Consider the G action on $k[G]$. Since $k[G]$ is noetherian, we can find $\gamma_1, \cdots, \gamma_r$ generate $k[G]$ as k-algebra. By local finiteness we can find $V \supseteq \langle \gamma_1, \cdots, \gamma_r \rangle$ for some finite dimensional G-stable subspace V. We only need to show G action is faithful on V. But this is obvious: if g fixes V then g fixes every γ_i, so g acts trivially on $k[G]$. But regular functions on affine varieties separate points, so G acts faithfully. □

Let's now recall some representation theory of finite groups. We know $k[G]$ is a representation of $G \times G$.

1. Suppose k is algebraically closed with $\text{char}k \nmid |G|$, then:
$$k[G] = \bigoplus_{V \text{irreducible}} V \otimes V^*$$
2. Suppose k arbitray, we have Frobenius Reciprocity: $\text{Hom}_G(k[G], V) = Hom_k(k, V) = V$, so any irreducible representation V occurs as a quotient of $k[G]$
3. Similarly, $\text{Hom}_G(V, k[G]) = \text{Hom}_k(V, k) = V^*$, so every irreducible representation occurs a submodule of $k[G]$.

Lemma 2.11. *Let G be an affine algebraic group. There is a unique irreducible component of G, say G°, containing identity. Moreover, G° is a finite index closed normal subgroup of G, and G° is the connected component of G containing the identity.*

Proof. Let X, Y be irreducible components containing identity. We know $\overline{XY}$, X^{-1} are both irreducible. This forces $X = \overline{XY} = X^{-1}$. Hence $X = Y$ is a closed subgroup of G. Hence such G° exists. Note then for each $g \in G$, we know $gG^\circ g^{-1}$ is an irreducible component containing identity, this forces $gG^\circ g^{-1} = G^\circ$ and G° is normal. It follows that G° is finite index since G has only finitely many irreducible components. Now we pick a coset representative A for the finite set G/G°, we then know topologically:

$$G = \coprod_{g \in A} gG^\circ$$

Hence G° is indeed the connected component containing the identity. □

Example 2.12. *Let G be an affine algebraic group, V a finite dimensional representation.*

1. $\mathrm{Hom}_G(k[G], V) \neq V$. *Consider $G = \mathbb{G}_a$, and $V = k, G$ acts on V via $a \cdot v = av$.*

   Claim: We have $\mathrm{Hom}_G(k[G], k) = 0$.

   Proof of Claim: Fix $\varphi : k[G] \to V$. Note $k[G]$ has a basis $\{1, x, x^2, \cdots\}$ We will show $\varphi(x^n) = 0$. For $n = 0$, note 1 is invariant under G-action, so $\varphi(1)$ invariant under G-action, which forces it to be 0 . Now we induct on n. Suppose the statement is true for $k \leq n-1$. We compute $a \cdot \varphi(x^n) = \varphi((x-a)^n) = \varphi(x^n)$, for any $a \in G \implies \varphi(x^n) = 0$

2. $\mathrm{Hom}_G(V, k[G]) = V^*$. *We have the following isomorphism*

   $$V^* \to \mathrm{Hom}_G(V, k[G]), \lambda \mapsto (v \mapsto \varphi(v) = (g \mapsto \lambda(g^{-1}v)))$$

   This is G-equivariant: $\varphi(hv)(g) = \lambda(g^{-1}hv) = \lambda((h^{-1}g)^{-1}v) = (h \cdot \varphi(v))(g)$.

   This is injective: $\varphi(v)(1)$ sees λ.

   This is surjective: Suppose I have $\theta : V \to k[G]$, define $\lambda(v) = \theta(v)(1)$, this is k-linear. We see then θ is induced by λ.

   As a remark, if V is irreducible, then V embeds into $k[G]$ and is finite-dimensional (since $k[G]$ is locally finite). Therefore, the category of G-representations has enough injectives.

Remark 2.13. *The existence of the distinguished representation $k[G]$ of G is remarkable. Note in the case of Lie algebra, there is not a distinguished locally finite left $\mathfrak{g}$-module: the universal enveloping algebra $U(\mathfrak{g})$ is free as a left $\mathfrak{g}$-module.*

2.2 Jordan Decomposition

Let V finite dimensional vector space over k, $\sigma : V \to V$ linear map.

Definition 2.14. We call σ semisimple if $V \otimes \overline{k} \to V \otimes \overline{k}$ is diagonalizable $\iff$ minimal polynomial of σ in $k[x]$ splits in $\overline{k}[x]$ splits into distinct linear factors. We call σ nilpotent if $\sigma^n = 0$ for some n. We call σ unipotent if $\sigma - 1$ nilpotent $\iff$ all eigenvalues of $\sigma \otimes \overline{k}$ are 1.

Remark 2.15. *σ semisimple and nilpotent $\implies \sigma = 0$.*

Lemma 2.16. *Any commuting set $\gamma_i : V \to V$ of endomorphisms of V can be simultaneously upper triangulized such that if γ_i is semisimple, then γ_i is diagonal.*

Proof. Induct on the cardinality of the set. □

Proposition 2.17. *Let $\sigma \in \mathrm{End}(V)$.*

1. *There exists $N, S \in \mathrm{End}(V \otimes \overline{k})$ such that $\sigma = S + N$ for S semisimple, N nilpotent, $SN = NS$.*
2. *These properties characterize S and N uniquely, and $S = p(\sigma)$ for $p(x) \in x\overline{k}[x]$.*
3. *if k is perfect, then $S, N \in \mathrm{End}(V)$.*

Proof. We only show the third part. Let $g \in \mathrm{Gal}(\overline{k}/k)$. Then $g\sigma = \sigma$, $g\sigma = g(S + N) = gS + gN$ for gS semisimple and gN nilpotent. Hence the uniqueness we just proved shows that $gS = S, gN = N$, so $S, N \in \mathrm{End}(V \otimes \overline{k})^{\mathrm{Gal}(\overline{k}/k)}$. If k is perfect, we know $\mathrm{End}(V \otimes \overline{k})^{\mathrm{Gal}(\overline{k}/k)} = \mathrm{End}(V)$. □

Remark 2.18. *The third statement is false if k is not perfect. And we will assume k is perfect shortly.*

Remark 2.19. *σ invertible $\iff$ S is invertible, and they have the same eigenvalues. Moreover: S^{-1} is a polynomial in σ.*

Proof. Let f be the minimal polynomial of σ and $S = p(\sigma)$. But σ is invertible implies $\gcd(p, (x - \lambda_i)^{n_i}) = 1$ as $p \equiv \lambda_i \bmod (x - \lambda_i)^{n_i}$. So $\gcd(p, f) = 1$, and we pick $a, b \in \overline{k}[x]$ with $1 = ap + bf$. So $I = a(\sigma)p(\sigma)$, $S^{-1} = a(\sigma)$. □

Corollary 2.20. *If $\sigma \in \mathrm{End}(V)$ is invertible, we can write $\sigma = su$, for s semisimple and u unipotent, $su = us$. And this decomposition is unique.*

Now let V be a vector space over k, $\sigma \in \mathrm{End}(V)$ be locally finite.

Definition 2.21. We call σ is semisimple / nilpotent /unipotent, if $V = \sum W_i$ for W_i finite dimensional and σ-stable and $\sigma|_{W_i}$ is semisimple / nilpotent /unipotent.

Exercise. Let $\sigma \in \mathrm{End}(V)$ be locally finite.

1. Show σ is invertible $\iff$ all eigenvalues of σ are non-zero σ is injective.
2. Deduce the previous propositions still hold if σ is locally finite.

Let G be an affine algebraic group over a perfect field k. We write $g \in G$ for $g \in G(k)$. Define $\rho_g : k[G] \to k[G], f(x) \mapsto f(xg)$; $\lambda_g : k[G] \to k[G], f(x) \mapsto f(g^{-1}x)$.

Definition 2.22. We call $g \in G$ semisimple / unipotent if ρ_g is.

Remark 2.23. *g is semisimple (resp. unipotent) if and only if g^{-1} is. Moreover $\lambda_g = \iota \circ \rho_g \circ \iota$ for $\iota(f) = \{g \mapsto f(g^{-1})\}$. So ρ_g is semisimple (resp. unipotent) if and only if λ_g is.*

Proposition 2.24 (Jordan Decomposition for Linear Algebraic Groups)**.** *Assume G is reduced and k is perfect. Let $g \in G(k)$. Then there exists unique $u, s \in G$ such that $g = su$ for u unipotent, s semisimple, $su = us$. We call this the Jordan Decomposition of $g \in G$.*

Proof. Write $\rho_g = SU$ for $S, U \in \mathrm{End}(k[G])$. We need to show $S = \rho_s, U = \rho_u$ for some $s, u \in G$. The proposition follows from several claims below.
Claim: $S, U \in \mathrm{End}(k[G])$ is an algebra automorphism. This follows from the following general lemma:

Lemma. *Let A be an algebra over k, $\sigma : A \to A$ k-algebra automorphism, σ locally finite. Let $\sigma = SU$ be Jordan decomposition. Then $S : A \to A$ is an algebra automorphism, hence so is $U = \sigma S^{-1}$.*

Proof of Claim: It suffices to show $S : A_{\overline{k}} \to A_{\overline{k}}$ is an algebra automorphism. Write $A_\lambda = \{x \in A : (\sigma - \lambda)^N = 0, \text{for some N} > 0\}$. Since σ locally finite, $A_{\overline{k}} = \bigoplus_{\lambda \in \overline{k}^*} A_\lambda$, and S acts as multiplication by λ on A_λ. We need to show $A_\lambda A_\mu \subseteq A_{\lambda\mu}$, but this is basic linear algebra.
Claim: U, S commutes with the left multiplication map λ_x for any $x \in G$.
Proof of Claim: Indeed, left and right multiplication commutes. But $k[G] = \cup V$ for V finite dimensional and $G \times G$-stable. Note that for any such V, $S|_V, U|_V$ are polynomials in ρ_g. So they commute with λ_x.
Claim: If $\alpha : k[G] \to k[G]$ is algebra automorphism, $k = \overline{k}$. Suppose further $\alpha\lambda_x = \lambda_x\alpha$ for any $x \in G$. Then there exists unique $h \in G$ such that $\alpha = \rho_h$.
Proof of Claim: Consider $k[G] \to k, f \mapsto \alpha(f)(1)$. This is an algebra map. But by Hilbert Nullstellensatz, we know $\alpha(f)(1) = f(h)$ for a unique $h \in G$. Now that $\alpha\lambda_x = \lambda_x\alpha \implies (\alpha(f))(x^{-1}) = f(x^{-1}h) \implies \alpha(f) = \rho_h(f) \implies \alpha = \rho_h$. The uniqueness if clear.

Note then this shows $S = \rho_s, U = \rho_u$ for unique $s, u \in G(\overline{k})$, for s semisimple and u unipotent. But since k is perfect, we have $s, u \in G(k) = G(\overline{k})^{\mathrm{Gal}(\overline{k}/k)}$. □

Proposition 2.25. *Let $\gamma : G \to \mathrm{GL}_V$ be a finite dimensional representation of G. Suppose $g = su$ is the Jordan decomposition of g. Then $\gamma(g) = \gamma(s)\gamma(u)$ is the standard Jordan decomposition of a linear map.*

Proof. It suffices to find an embedding $V \hookrightarrow k[G]^{\oplus N}$ for some N (i.e. any finite dimensional representation has an injective resolution). Indeed, recall we've constructed the isomorphism:

$$V^* \to \mathrm{Hom}_G(V, k[G]), \lambda \mapsto (v \mapsto \varphi(v) = (g \mapsto \lambda(g^{-1}v)))$$

Choose a basis $e_1^*, \cdots, e_n^*$ of V^*. One can check that $v \mapsto (g \mapsto (e_1^*(g^{-1}v), \cdots, e_n^*(g^{-1}v)))$ is an G-equivariant embedding. □

Remark 2.26.
1. *If $\mathfrak{g} = T_I G$, then the Jordan decomposition in G gives rise to Jordan decomposition in $\mathfrak{g}$. But this isn't necessarily true for "non-algebraic" Lie algebras.*
2. *The natural map $G(k) \to G(k), g \mapsto g_s$ is not a morphism of algebraic varieties. For $G = \mathrm{SL}_2$, we know $G^{ss} = \{g \in G | \mathrm{tr}(g) \neq \pm 2\} \cup \{\pm I\}$ is dense in G. So $g \mapsto g_s$ is identity on a dense subset of G, which forces this map to be identity if the natural map is in fact a morphism, contradiction.*

Example 2.27. *Let G be finite, and k be of characteristic p. We know $g \in G$ is semisimple $\iff$ $p \nmid ord(g)$; $g \in G$ is unipotent $\iff$ $ord(g) = p^N$ for some N.*

Exercise. Let $g = su \in G$ be Jordan decomposition. Then:

$$Z_G(g) = Z_{Z_G(s)}(u) = Z_G(s) \cap Z_G(u).$$

3 Quotients

Let G be affine algebraic group, $H \leq G$ a closed subgroup. In what sense G/H is an algebraic variety? We will construct G/H as G-orbit of a point in a projective space $\mathbb{P}(V)$ (more precisely, we will construct a G-representation V, and $x \in \mathbb{P}(V)$ such that $\mathrm{stab}_G(x) = H$). We will also see this construction really behaves like a "quotient".

More generally, let G be an algebraic group acting on variety X. We want to give "the orbit of G on x" an algebraic variety structure. Let $X/_{\text{set}}G = X(k)/G(k)$ be the set of orbits of $G(k)$ on $X(k)$. There are three ways to deal with the problem: sheafify, geometric invariant theory, algebraic stacks.

Assume for simplicity $k = \overline{k}$.

Lemma 3.1. *Suppose an affine algebraic group G acts on quasi-projective variety X. Then the G-orbit $G \cdot x$ is a locally closed subvariety (i.e. open in its closure). Moreover, $\overline{G \cdot x}$ is a union of G-orbits. If $y \in \overline{G \cdot x} \backslash G \cdot x$, then $\dim G \cdot y < \dim G \cdot x$.*

Proof. We have the following general statement from algebraic geometry:
Claim: Let $\alpha : X \to Y$ be morphism of varieties. Then $\alpha(X)$ contains a nonempty open subset of $\overline{\alpha(X)}$.

Apply this to $\alpha : G \to X, g \mapsto gx$, we see $G \cdot x$ contains $\emptyset \neq U$ open in $\overline{G \cdot x}$. But then $G \cdot x = \cup_{g \in G}(g \cdot U)$ is open in $\overline{G \cdot x}$. This shows $G \cdot x$ is locally closed. The fact that $\overline{G \cdot x}$ is a union of G-orbits is a purely topological statement. Moreover, the statement on the dimension follows from the following general claim for varieties: for X irreducible variety, U open in X, then $\dim(X \backslash U) < \dim X$. □

Example 3.2. *G acts on G by conjugation. Therefore, conjugate classes of G are locally closed, the closure is a union of conjugacy classes of smaller dimensions.*

Corollary 3.3. *Let G acts on X. The orbits of minimal dimension are closed. In particular, closed orbits exist.*

Lemma 3.4. *Let $\varphi : H \to G$ be a morphism of affine algebraic groups.*

1. *$\varphi(H)$ is closed in G.*
2. *$\varphi(H^\circ) = \varphi(H)^\circ$.*
3. *$\dim H = \dim \ker \varphi + \dim \mathrm{im}\ \varphi$*

Proof. 1. Let H act on G by left multiplication, the orbits are all isomorphic to $\varphi(H)$, hence in particular have the same dimension. This shows all H-orbits of G are closed. In particular $\varphi(H)$ is closed.

2. $\varphi(H)$ is an algebraic group containing an irreducible subgroup $\varphi(H^\circ)$. So $\varphi(H^\circ) \subset \varphi(H)^\circ$. But $\varphi(H^\circ)$ is a closed subgroup of finite index in $\varphi(H)^\circ$, so it is open with open complement. So $\varphi(H^\circ) = \varphi(H)^\circ$.

3. Recall for $\alpha : X \to Y$ morphism and X irreducible, $\alpha(X) = Y$. Then $\dim \alpha^{-1}(y) \geq \dim X - \dim Y$, for any $y \in Y$, and is equality if y is in a dense open subset of Y. The statement follows immediate by considering $\varphi : H \to \varphi(H)$.

□

Proposition 3.5. *Let G be an affine algebraic group, with closed subgroup H. Then there exists finite dimensional representation V of G and a line $l \in \mathbb{P}(V)$ such that* $\mathrm{stab}(l) = H$.

Proof. Consider $I = \{f \in k[G] : f|_H = 0\}$ be the ideal of $k[G]$ defining the subgroup H. Since $k[G]$ is noetherian, we pick a generating set $\{\gamma_1, \cdots, \gamma_r\}$ of I, and take V_0 be the G-span of them. Since the G-action on $k[G]$ is locally finite, so $\dim V_0 < \infty$. We let $W = V_0 \cap I$ be a k-vector space containing γ_i for each i. We see $W \cdot k[G] = I$.

We let ${}^h f = \lambda_h f$, and so $f \mapsto {}^h f$ is an algebra automorphism. Note that ${}^h I = I$ for each $h \in H$, so $H \leq \{g \in G, {}^g W = W\}$. But if ${}^g W = W \implies {}^g I = {}^g(W \cdot k[G]) = {}^g W \cdot {}^g k[G] = W \cdot k[G] = I$. But gH is the zero set of ${}^g I \implies gH = H \iff g \in H$. So $H = \{g \in G, {}^g W = W\}$.

We now want to view W as a point of a projective space. Note G acts on V_0, hence acts on $\mathrm{Gr}_d(V_0), d = \dim W$. Now we view $W \in \mathrm{Gr}_d(V_0)$, then $H = \mathrm{stab}_G(W)$. We then embed the Grassmannian into a projective space $\mathbb{P}(\bigwedge^d V_0)$ via the Plucker embedding. Consider $l = \bigwedge^d W \in \mathbb{P}(\bigwedge^d V_0)$ to prove the proposition. □

Corollary 3.6. *We can then define $G/H \coloneqq G \cdot l$. One sees that G/H is indeed a quasi-projective variety, and $G \cdot l(k) = G/_{set} H$.*

So if $H \leq G$, we can realize the quotient G/H as an orbit $G \cdot x$ for some $x \in \mathbb{P}V$. A natural question to ask is: if $y \in \mathbb{P}V'$ such that $\mathrm{stab}_G(y) = H$, is it true that $G \cdot x \cong G \cdot y$ as varieties? More generally, if G is an affine group acting on a variety X, in what sense the k-points of the "orbits of G on X" forms an algebraic variety? We will sketch the idea of Geometric Invariant Theory (GIT) through a few examples.

Assume we're working in algebraically closed field k. Let X be an affine variety, and G is a reduced affine algebraic group acting on X. Consider $k[X]^G$ of G-invariant functions on X (i.e. each $f \in k[X]^G$ is constant on each fibre).

The inclusion map $k[X]^G \hookrightarrow k[X]$ induces:

$$x \longmapsto \{f \in k[X]^G \mid f(x) = 0\}$$

$$\begin{array}{ccc} X & \longrightarrow & \mathrm{Spec}\ k[X]^G \\ & \searrow \quad \nearrow & \\ & X/_{set} G & \end{array}$$

There are a few natural qustions to ask:

1. Do invariant functions separate orbits? (i.e. is $X/_{\text{set}}G \to \text{Spec } k[X]^G$ injective?)
2. Is $X/_{\text{set}}G \to \text{Spec } k[X]^G$ surjective?
3. When is $k[X]^G$ finitely generated? (this is known as "Hilbert's 14-th problem").

Example 3.7. *1. Let $X = \mathbb{A}^2, G = \mathbb{G}_m, t \cdot (x, y) = (tx, ty)$, then $k[X]^{\mathbb{G}_m} = k$. But X/G should be viewed as $\mathbb{P}^1 \cup \{pt\}$, where the point is in the closure of any other point. We can make sense of it using the language of stack.*

2. Let $X = \mathbb{A}^2, G = \mathbb{G}_m, t \cdot (x, y) = (tx, t^{-1}y)$, then $k[X]^{\mathbb{G}_m} = k[xy]$. Let:

$$p : \mathbb{A}^2/_{set}\mathbb{G}_m \to \text{Spec } k[xy]$$
$$(x, y) \mapsto xy$$

So

$$p^{-1}(z) = \begin{cases} \textit{a single orbit}, & \textit{if } z \neq 0; \\ \textit{three orbits}, & \textit{if } z = 0. \end{cases}$$

Note that if $\mathcal{O} \subset X$ is an orbit, we know $\overline{\mathcal{O}}$ is a union of orbits. Now if $f \in k[X]$ vanishing on $\mathcal{O}$, then it must vanish on $\overline{\mathcal{O}}$. In particular $f(\mathcal{O}') = 0$ for any $\mathcal{O}' \subset \overline{\mathcal{O}}$. So $k[X]^G$ cannot distinguish $\mathcal{O}$ from an orbits in $\overline{\mathcal{O}}$. However, we see in the previous two examples, $k[X]^G$ indeed parametrizes ***closed orbits****.*

The idea of GIT is that if (G, X) is "nice", then $k[X]^G$ encodes an algebraic geometric structure of the closed points of G on X. This induces an algebraic geometric structure on the equivalence classes of all orbits (the equivalence relation is generated by $\mathcal{O}_1 \sim \mathcal{O}_2$ if $\overline{\mathcal{O}_1} \cap \mathcal{O}_2 \neq \emptyset$).

Exercise. Let $X = \text{SL}_2 = G$, G acts on X by conjugation.

1. a conjugacy class ${}^G g$ is closed $\iff$ g is semisimple.
2. every semisimple conjugacy class intersects $T = \left\{ \begin{pmatrix} t & \\ & t^{-1} \end{pmatrix} \right\}$ in a $N(T)/T = S_2$
3. Show the following isomorphisms:

$$\text{Spec } k[X]^G \xrightarrow{\text{res}} \text{Spec } k[T]^{S_2} \xrightarrow{t \mapsto t + t^{-1}} \mathbb{A}^1, \quad g \mapsto \text{tr}(g)$$

This shows { closed orbits = semisimple orbits} = { orbits of S_2 in $\mathbb{G}_m$ } = $\mathbb{A}^1$

This leads to the following guess: For a semisimple algebraic group G, $k[G]^G \cong k[T]^W$. Moreover, GIT also requires the pair (G, X) to be nice. The pair (affine variety, reductive group) is nice, while the pair (affine variety, unipotent group) is not.

Example 3.8. *1. $X = \mathbb{A}^4, G = \mathbb{G}_a$, let $s \cdot (x_1, x_2, y_1, y_2) = (x_1 + sy_1, x_2 + sy_2, y_1, y_2)$. Then $k[X]^G = k[y_1, y_2, x_1y_2 - x_2y_1 = a]$, so we get $X/G \to \mathbb{A}^3$ via $(x_1, x_2, y_1, y_2) \mapsto (y_1, y_2, x_1y_2 - x_2y_1)$ is not surjective.*

2. Let $G = \mathbb{G}_a^3, X = \mathbb{A}^{18}$. Nagata, Steinberg and Mukai constructed an action of G on X such that $k[X]^G$ is not finitely generated.

3. Let $G = X = \mathrm{SL}_2$, $T = \left\{ \begin{pmatrix} a & \\ & a^{-1} \end{pmatrix} \right\}$ acting on X be right multiplication:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} at & bt^{-1} \\ ct & dt^{-1} \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} t & \\ & t^{-1} \end{pmatrix}$$

One sees that $k[X]^T = k[ab, cd, ad, bc] = k[\alpha, \beta, \gamma, \delta]/(\alpha\delta = \beta\gamma, \beta - \gamma = 1)$

On the other hand, choose $x \in X$ such that $\mathrm{tr}(x) = \lambda \neq \pm 2$. Then x is conjugate to a diagonal matrix $\pm 1 \neq t \in T$. One checks that $Z_G(t) = T$. So we know x belongs to orbit $\mathcal{O}_\lambda = \{x \in X | \mathrm{tr}(x) = \lambda\}$. Now SL_2 acts transitively on $\mathcal{O}_\lambda$, with stablizer of t equals T. So: $\mathrm{SL}_2/_{set}T \simeq \mathcal{O}_\lambda$.

Evidently, $k[\mathcal{O}_\lambda] \cong k[X]^T$. Therefore, each semismiple conjugacy class itself is a homogeneous space. Moreover, if $\lambda \neq \pm 1$, then it equals (note $\mathcal{O}_\lambda$ is affine)

$$\mathrm{SL}_2/_{set}T \cong \mathcal{O}_\lambda = \mathrm{Spec}\ k[\mathcal{O}_\lambda] \cong \mathrm{Spec}\ k[\mathrm{SL}_2]^T$$

Compare with our previous result:

$$\{\textit{semisimple conjugacy classes}\} = T/\pm 1 = G^{ss}/_{set}G = \mathrm{Spec}\ k[G]^G = \mathbb{A}^1$$

4. Let $X = \mathrm{SL}_2$, $B = \left\{ \begin{pmatrix} * & * \\ & * \end{pmatrix} \right\}$ be a Borel, acting by right multiplication. We've seen $\mathrm{SL}_2/_{set}B \cong \mathbb{P}^1(k)$ as a set, and $k[\mathrm{SL}_2]^B = k = \Gamma(\mathbb{P}^1, \mathcal{O})$.

So we have a "fibration": $p : \mathrm{SL}_2 \to \mathbb{P}^1$, $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \frac{a}{c} \in \mathbb{P}^1$. Observe that:

$$p^{-1}(\mathbb{P}^1 \backslash \{\infty\}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} | c \neq 0, ad - bc = 1 \right\} = \left\{ \begin{pmatrix} 1 & b \\ & 1 \end{pmatrix} \begin{pmatrix} & 1 \\ -1 & \end{pmatrix} \begin{pmatrix} c & x \\ & c^{-1} \end{pmatrix} \right\} = \mathbb{A}^1 \times B$$

*This is the **Bruhat Decomposition**. Note this identification is B-equivariant. Hence for any $z_0 \in \mathbb{P}^1$, $p^{-1}(\mathbb{P}^1 \backslash \{z_0\}) = V_{z_0}$ is an affine variety equipped with an equivariant B action: $V_{z_0} \cong \mathbb{A}^1 \times B$.*

Hence $k[V_{z_0}]^B = k[\mathbb{P}^1 \backslash \{z_0\}] = k[t]$. This justifies that p is indeed a fibration in the Zariski topology.

Definition 3.9. Let G acts on X freely and trivially on Y. We call X a **principal G-bundle** or **princpal G-torsor** on Y in *Zariski topology*, if there is a G-equivariant map $p : X \to Y$ such that:

1. for $U \subset Y$ Zariski open, $\Gamma(U, \mathcal{O}) = \Gamma(p^{-1}U, \mathcal{O})^G$
2. for any $y \in Y$, there is a Zariski affine open nieghborhood U, such that $p^{-1}U \cong U \times G$ equivariantly.

Definition 3.10. Let G acts on X freely and trivially on Y. We call a G-equivariant map $p : X \to Y$ a **flat G-quotient** if

1. for $U \subset Y$ Zariski open, $\Gamma(U, \mathcal{O}) = \Gamma(p^{-1}U, \mathcal{O})^G$
2. each fibre is isomorphic to G as an algberaic variety.
3. $p : X \to Y$ is flat.

Remark 3.11. *1. A principal G-bundle in Zariski topology forms a flat G-quotient.*

2. If p is a flat G-quotient, then it is open (since p is finite type), and affine (by flat base change and Serre's criterion).

Exercise. Show that if the above properties hold on some affine cover, then they hold on all affine opens.

Corollary 3.12. *$p : \mathrm{SL}_2 \to \mathbb{P}^1$ is a principal B-bundle in Zariski topology.*

In order to define $Y = X/G$ satisfying the first property (in both definitions), we need to sheafify the functor $X \to \operatorname{Spec} k[X]^G$. We need to find enough G-invariant open affine $V = \operatorname{Spec} A$, with $V/_{\mathrm{set}}G \simeq \operatorname{Spec} A^G$ covering X. Then we can glue these $\operatorname{Spec} A^G$ to get the quotient X/G (easy to check the gluing condition).

Example 3.13. *Let $X = \mathbb{G}_m, G = \mathbb{Z}/2\mathbb{Z}$, acting by $t \mapsto -t$. Consider the map:*

$$\mathbb{G}_m \to \mathbb{G}_m, t \mapsto t^2$$

We compute $k[X]^G = k[t, t^{-1}]^{\mathbb{Z}/2\mathbb{Z}} = k[t^2, t^{-2}]$. Note as a $k[t^{\pm 2}]$-module, $k[t^{\pm 1}] \cong k[t^{\pm 2}] \otimes \langle 1, t \rangle$. This is not a Zariski locally trivial fibration by examining the connectedness. Note that for Zariski locally trivial fibration, we have $k[p^{-1}U] = k[U] \otimes k[G]$ as algebras for U small enough; but here $k[p^{-1}U] = k[U] \otimes k[G]$ just as $(k[U], G)$-module.

Lemma 3.14. *Let $p : X \to Y$ be a flat G-quotient, then locally (on a Zariski open neighborhood) on Y: $p_*\mathcal{O} = \mathcal{O}_Y \otimes k[G]$ as a sheaf of G-modules.*

Proof. This follows immediately from the next lemma. □

Lemma 3.15. *Let $R \to S$ be an inclusion of k-algebras, such that S/R is flat and finite type. Assume R is reduced. Suppose G acts on $\operatorname{Spec} S$, and acts trivially on $\operatorname{Spec} R$. Suppose for any $\mathfrak{p} \in \operatorname{Spec} R$ such that $S/\mathfrak{p}S := S \otimes_R k_{\mathfrak{p}} \cong k[G]$ as G-modules. Then for any $\mathfrak{p} \in \operatorname{Spec} R$, there exists $f \in R$ such that for any $\mathfrak{p} \in D(f)$, $S_f \cong R_f \otimes k[G]$ as a $R_f \otimes k[G]$-module, where $R_f = R[\frac{1}{f}]$.*

Remark 3.16. *Let's unwrap what does it mean for a G-action on $\operatorname{Spec} S$. There is a map $G \times \operatorname{Spec} S \to \operatorname{Spec} S$, which is the same as a $k[G]$-comodule structure on S: $S \to S \otimes k[G]$ $\iff$ G acts on S by algebra automorphism.*

Proposition 3.17. *Given condition 3 and 4, it suffices to check 1 for an affine cover.*

Proof. *Claim:* Suppose U is open affine in Y and $\Gamma(p^{-1}U,\mathcal{O})^G = \Gamma(U,\mathcal{O})$, then for any $V \subset U$, we have $\Gamma(p^{-1}V,\mathcal{O})^G = \Gamma(V,\mathcal{O})$.
Proof of Claim: Let $f \in \Gamma(U,\mathcal{O}), U_f = \{u \in U, f(u) \neq 0\}$. Then $\Gamma(U_f,\mathcal{O}) = \Gamma(U,\mathcal{O})[\frac{1}{f}] = (\Gamma(p^{-1}U,\mathcal{O})[\frac{1}{f}])^G = \Gamma(p^{-1}U_f,\mathcal{O})^G$. So the claim holds for U_f, we can then see it holds for all open set V by gluing.

Now suppose $V = \cup U_i$, and $\Gamma(p^{-1}U_i,\mathcal{O})^G = \Gamma(U_i,\mathcal{O})$. We want to show $\Gamma(V,\mathcal{O}) = \Gamma(p^{-1}V,\mathcal{O})^G$. Indeed, we have a Cech Cover:

$$V \longleftarrow \coprod U = U \longleftarrow \coprod U_i \cap U_j$$

$$0 \longrightarrow \Gamma(V,\mathcal{O}) \longrightarrow \Gamma(U,\mathcal{O}) \longrightarrow \Gamma(\coprod U_i \cap U_j,\mathcal{O})$$

As p is affine, we have

$$\Gamma(p^{-1}V,\mathcal{O}) = \ker\{\Gamma(p^{-1}U,\mathcal{O}) \to \Gamma(\coprod p^{-1}(U_i \cap U_j),\mathcal{O})\}$$

But if we set $A = \Gamma(p^{-1}U,\mathcal{O}), B = \Gamma(\coprod p^{-1}(U_i \cap U_j),\mathcal{O})$,then:

$$\Gamma(p^{-1}U,\mathcal{O})^G = \Gamma(U,\mathcal{O}) \ ,\Gamma(\coprod p^{-1}(U_i \cap U_j),\mathcal{O})^G = \Gamma(\coprod U_i \cap U_j,\mathcal{O})$$

by the previous claim. So:

$$\Gamma(p^{-1}V,\mathcal{O}) = \ker(A \to B) \implies \Gamma(p^{-1}V,\mathcal{O})^G = \ker(A \to B)^G = \ker(A^G \to B^G) = \Gamma(V,\mathcal{O})$$

. □

Remark 3.18. *In fact if $p : X \to Y$ is a flat G-quotient, then:*

$$H^i(Y,\mathcal{O}) = H^i(X,\mathcal{O})^G$$

, since $\Gamma(p^{-1}(U_1 \cap \cdots \cap U_i),\mathcal{O}) = \Gamma(U_1 \cap \cdots \cap U_i,\mathcal{O}) \otimes k[G]$, so taking G-invariant is exact. In particular, this shows if Y is a flag variety, then $H^i(Y,\mathcal{O}) = 0$ for each $i > 0$.

If G acts freely on X affine, $Y = \operatorname{Spec} k[X]^G$ is a flat G-quotient $\iff$ if $k[X]$ is a flat $k[X]^G$-module, and fibres are all isomorphic to G.

Example 3.19. *1. Show this happens if G is a finite group acting freely on affine X.*

2. *Recall that open immersions, smooth morphisms are flat ($f : X \to Y$ smooth $\implies df : TX \to f^*TY$ is a quotient of vector bundles).*
3. *$k[t^n] \to k[t]$ is flat.*

4. *Let $\mu_n(k)$ be n-th roots of unity in k. So $\mu_n(k) = k[t]/(t^n - 1) \subset \mathbb{G}_m$ is a closed subgroup. We can construct the quotient $\mathbb{G}_m/\mu_n(k)$. Now consider the action from $\mathbb{G}_m$ to $\mathbb{G}_m$ by letting a acting as multiplication by a^n. Then the orbit of 1 is the whole $\mathbb{G}_m$, with stablizer $\mu_n(k)$.*

$$0 \to \mu_n \to \mathbb{G}_m \to \mathbb{G}_m \to 0$$

Check this is a flat quotient. Note if $n = p$, then the "quotient" is identity on the level of $\overline{k}$-points.

Exercise. 1. Let $\mathrm{PGL}_n = \mathrm{GL}_n/\mathbb{G}_m$ is an affine algebraic group by writing its defining equations. Note $\mathrm{PGL}_n \subset (\mathrm{Mat}_n \backslash \{0\})/\mathbb{G}_m = \mathbb{P}^{n^2-1}$ is a quasi-projective variety given by nonvanishing determinant. Also, show that the orbit construction is really a quotient.

2. Show $\mathrm{PSL}_n = \mathrm{SL}_n/\mu_n$ is an affine algebraic group. So we have:

$$\begin{array}{ccccc}
\mu_n & \hookrightarrow & \mathbb{G}_m & \xrightarrow{det} & \mathbb{G}_m \\
\downarrow & & \downarrow & & \downarrow \\
\mathrm{SL}_n & \hookrightarrow & \mathrm{GL}_n & \xrightarrow{det} & \mathbb{G}_m \\
\downarrow & & \downarrow & & \\
\mathrm{PSL}_n & \longrightarrow & \mathrm{PGL}_n & &
\end{array}$$

Show that $\mathrm{PSL}_n \cong \mathrm{PGL}_n$.

Remark 3.20. *Note the above exercise depends on the fact that k is algebraically closed. And care needs to be taken for $n = 0$ in k. Moreover, it worths noticing that $(G/H)(k) \neq G(k)/H(k)$.*

Example 3.21. *Note $\mathrm{PSL}_2(\mathbb{F}_p) = \mathrm{SL}_2(\mathbb{F}_p)/(\pm 1)$, $\mathrm{PGL}_2(\mathbb{F}_p) = \mathrm{GL}_2(\mathbb{F}_p)/\mathbb{F}_p^\times$. Note the later contains $\mathrm{diag}(1, -1)$, but not in the former if -1 is a quadratic non-residue modulo p.*

Grothendiek's picture: a scheme X/k is determined by its A-point for A algebra over k. $X(A) = \mathrm{Hom}(\mathrm{Spec}\, A, X)$, and it equals $\mathrm{Hom}_{\mathrm{alg}}(k[X], A)$ if X affine. In particular, if $\varphi : Z \to Z'$ is a morphism of schemes, φ is an isomorphism if $Z(A) \to Z'(A)$ is a bijection of sets for all A.

This raises the problem: how do we think of $G/H(A)$ as it does not equal $G(A)/H(A)$.

Proposition 3.22. *Let G act freely on X, trivially on Y and $p : X \to Y$ a G-equivariant morphism. Then $p : X \to Y$ is a flat G-quotient $\iff$ p is flat, surjective, and $G \times X \to X \times_Y X, (g, x) \to (gx, x)$ is an isomorphism. ("G is a locally trivial fibration in the flat topology".)*

Remark 3.23. *$X \times_Z Y(T) = X(T) \times_{Z(T)} Y(T)$ for all X, Y, Z, T schemes. So the property $G \times X \to X \times_Y X$ being an isomorphism can be checked pointwise.*

Proof. If $U \subset Y$ any subvariety, $V = p^{-1}(U)$, then:

$$\begin{array}{ccc} G \times V & \xrightarrow{\sim} & V \times_Y V = V \times_U V \\ \downarrow & & \downarrow \\ G \times X & \xrightarrow{\sim} & X \times_Y X \end{array}$$

is a pullback square as G-spaces. So by taking $U = \{y\}$, we immediately have $p^{-1}(y) \cong G$ as G-spaces.

Now if U open, $G \times V \cong V \times_Y V$.

Exercise. Conversely, show that if $G \times_Y p^{-1}(U) \cong p^{-1}(U) \times_Y p^{-1}(U)$ for U in some fixed affine cover of Y. Then $G \times_Y X \cong X \times_Y X$ is indeed an isomorphism.

Therefore, it suffices to check for the case for Y affine. Then one can also assume X affine (work needs to be done for RHS).

$$Y \xleftarrow{p} X \;\substack{\longleftarrow \\ \longleftarrow}\; G \times X \qquad\qquad k[Y] \xrightarrow{p^*} k[X] \;\substack{\longrightarrow \\ \longrightarrow}\; k[G \times X]$$

$$\begin{matrix} gx \\ x \end{matrix} \;\substack{\longleftarrow \\ \longleftarrow}\; (g, x) \qquad\qquad f \longmapsto \begin{matrix} (g,x) \mapsto f(gx) \\ (g,x) \mapsto f(x) \end{matrix}$$

Note G acts trivially on Y, so p^* factors through $\alpha : k[Y] \to k[X]^G$. The definition of the quotient is precisely the condition that α is an isomorphism.

On the other hand, we have:

$$Y \longleftarrow X \;\substack{\longleftarrow \\ \longleftarrow}\; X \times_Y X \qquad\qquad k[Y] \longrightarrow k[X] \;\substack{\longrightarrow \\ \longrightarrow}\; k[X] \otimes_{k[Y]} k[X]$$

$$\begin{matrix} x_1 \\ x_2 \end{matrix} \longmapsto (x_1, x_2) \qquad\qquad f \longmapsto \begin{matrix} f \otimes 1 \\ 1 \otimes f \end{matrix}$$

Theorem (faithfully flat descent)**.** *If $R \to S$ is a faithfully flat map of rings (i.e.* Spec $S \to$ Spec R *surjective, and $R \to S$ flat). Then the following sequence is exact:*

$$R \to S \to S \otimes_R S$$

where the second map is $s \mapsto s \otimes 1 - 1 \otimes s$.

So if $X \to Y$ surjective and flat, then $k[Y] \to k[X]$ is faithfully flat. So $k[Y] = \ker(k[X] \to k[X \times_Y X]) = \ker(k[X] \to k[G \times X]) = k[X]^G$. This proves the direction " $\Longleftarrow$ ". Conversely, assume Y is a flat G-quotient, then we know an isomorphism of G-modules:

$$k[X] = k[Y] \otimes k[G]$$

Hence we can compute:

$$k[X] \otimes_{k[Y]} k[X] \simeq k[G] \otimes k[X]$$

This gives us our desired isomorphism $G \times X \to X \times_Y X$. □

Remark 3.24. *This is an aside on faithfully flat descent. Suppose $Y = \cup U_i$ of open sets, $U = \coprod U_i$. Then $U \to Y$ is a faithfully flat map. Then $U \times_Y U = \coprod_{i,j} U_i \times_Y U_j = \coprod_{i,j} U_i \cap U_j$. And we have:*

$$Y \leftarrow U = \coprod U_i \leftleftarrows U \times_Y U = \coprod_{i,j}(U_i \cap U_j) \;\substack{\longleftarrow\\ \longleftarrow\\ \longleftarrow}\; U \times_Y U \times_Y U \cdots\cdots$$

The extra terms don't affect the kernel on the level of sections. Using "simplicial" sets we recover the standard Cech Complex.

Faithfully flat descent is R, S as above:

$$R \to S \to S \otimes_R S \to S \otimes_R S \otimes_R S \to \cdots$$

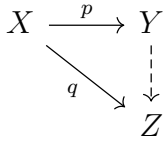
*This is a resolution called the **bar complex**. A special case is that the Cech cohomology doesn't depends on the affine cover. Another special case is $R = k$, and S is a k-algebra.*

Exercise. $G \times X \cong X \times_Y X$ can be checked pointwise. Moreover, the proposition implies $X(A)/G(A) \hookrightarrow Y(A)$.

Observe if $\underline{X}$ is a finite set, $\underline{G}$ a finite group acts freely on $\underline{X}$. Let $\underline{Y} = \underline{X}/\underline{G}$. Then $G \times X \to X \times_Y X, (g, x) \to (gx, x)$ is an isomorphism of sets. So checking the condition on A-points are easy.

Proposition 3.25. *If $p : X \to Y$ is a flat G-quotient, then p satisfies the following universal property:*

if $q : X \to Z$ is a morphism such that the fibres of q are union of G-orbits, then q factors through p.



In particular, if such a Y exists, then it is unique and written as X/G.

To prove our main theorem, we need the following result from algebraic geometry:

Theorem 3.26 (generic flatness)**.** *1. Let $R \hookrightarrow S$ be inclusion of finite type, for R reduced. Then there exists a dense open $U \subset \operatorname{Spec} R$ such that for any $y \in U$, there exists $f \in R$ such that $f(y) \neq 0$ such that S_f is a finitely presented R_f-algebra, which is free as a R_f-module.*

2. Let $p : X \to Y$ be morphism of schemes, Y reduced, p finite type. Then there exists a dense open $U \subset Y$ such that $X_U = p^{-1}(U) \to U$ is finitely presented and flat.

Proof. See Stack Project. □

Theorem 3.27. *1. Let $H \leq G$ closed subgroup, for G affine reduced. Let G acts on X, $x \in X(\overline{k})$ and x has stablizer H. Then the orbit map $G \to G \cdot x$ is a flat H-quotient. Moreover, $G \cdot x$ is smooth and locally closed subvariety of X , and $\dim G = \dim H + \dim(G/H)$. In particular $(G \cdot x)(\overline{k}) = G(\overline{k})/H(\overline{k})$.*

2. The following are equivalent:

(a) H is reduced

(b) If $\alpha : G \to G \cdot x$, $d\alpha_I : \mathfrak{g} = T_I G \to T_x(G \cdot x) = T_H G/H$ is surjective.

(c) α is a separable map, $\dim \mathfrak{g} = \dim \mathfrak{h} + \dim T_H G/H$

(d) $\ker(d\alpha_I) \subseteq \mathfrak{h}$

Proof. For any $\alpha : X \to Y$, $\mathrm{im}(\alpha)$ is a reduced constructible set provided X is. Now $G \cdot x$ is a homogeneous G-space, hence $G \cdot x$ is locally closed and smooth subvariety of X.

By generic flatness, there is an open set of $G \cdot x$ where the orbit map is flat. But G acts transitively, we see that $G \to G \cdot x$ is surjective and flat. Therefore, it remains to show that $G \to G \cdot x$ is H-torsor:

$$H \times G \simeq G \times_{G \cdot x} G$$

as G-spaces. However, this can be checked on A-points for any A:

$$G \times_{G \cdot x} G = \{(g_1, g_2) : g_1 x = g_2 x\} = \{(g_2^{-1} g_1, g_2) \in \mathrm{stab}_G(x) \times G\} = H \times G$$

□

Corollary 3.28. *Let $\alpha : G \to G'$ be a morphism between affine algebraic groups. Then $\ker \alpha$ is an affine normal subgroup, and $G/\ker \alpha \simeq \mathrm{im}(\alpha)$.*

Proposition 3.29. *Let $N \leq G$ be a closed normal subgroup. Then G/N is affine algebraic group.*

Proof. It suffices to find a representation W of G such that $N = \ker(p : G \to \mathrm{PGL}_W)$. Indeed, if so then $G/N \simeq \mathrm{im}(p) \subset \mathrm{PGL}_W$ is affine.

Recall that we can find a representation V of G, and $l \in \mathbb{P}V$ such that $\mathrm{stab}_G(l) = N$. Now let W be the smallest G-representation containing l: $W = G \cdot v, 0 \neq v \in l$. Note that N fixes l by construction, so we have:

$$\chi : N \to \mathbb{G}_m,\ n \cdot v = \chi(n) \cdot v \text{ for all } n \in N, v \in l.$$

But note that for $n \in N, g \in G$:

$$n \cdot (g \cdot v) = g \cdot (g^{-1} n g) \cdot v = {}^g\chi(n)(g \cdot v)$$

So N acts on $g \cdot l$ by ${}^g\chi$. Hence we can decompose W into isotypical N-spaces:

$$W = \bigoplus_{i=1}^{r} W_{\chi_i}, \text{ and that } \{\chi_i : 1 \leq i \leq r\} = \{{}^g\chi | g \in G\}$$

We select g_i such that ${}^{g_i}\chi = \chi_i$.
Claim: $T = \{g \in G | {}^g\chi = \chi\}$ is a closed subvariety of G.

But we then have $G = \bigcup_{i=1}^r g_i^{-1} T g_i$. We assume G is connected, then this forces $r = 1$ and $T = G$. This shows that:

$$N = \ker(p : G \to \mathrm{PGL}_W)$$

we're therefore done if we can deduce the general case for G not connected. Indeed: we've shown that $G^\circ/(G^\circ \cap N)$ is affine $\implies$ $G/(G^\circ \cap N)$ is affine since it's just disjoint union of copies of $G^\circ/(G^\circ \cap N)$. But then

$$G/N = (G/G^\circ \cap N)/(N/G^\circ \cap N)$$

is a quotient of affine group by a finite group, which remains affine. □

Remark 3.30. *The standard isomorphism theorems hold in the category of affine group schemes, but genuinely require proofs.*

4 Flag Varieties

4.1 Borel Subgroups and Flag Variety

Let G be a reduced affine connected algebraic group.

Definition 4.1. Call $B \subset G$ a **Borel Subgroup** if it's a maximal solvable connected reduced subgroup of G.

Remark 4.2.
1. *Since closure of connected reduced solvable subgroup still has these properties, Borel subgroups are closed.*
2. *Suppose $H_1 < H_2$ are connected reduced solvable, then $\dim H_1 < \dim H_2$. So every chain of solvable connected reduced subgroups stablizes.*

Theorem 4.3. *Let $k = \overline{k}$.*
1. *All Borel subgroups are conjugate.*
2. *Let B be any Borel subgroup. Then the flag variety G/B is projective.*

Remark 4.4. *The equations defining $G/B \hookrightarrow \mathbb{P}^N$ are very nice. The key word here is "Koszul".*

Corollary 4.5. *Non-trivial closed connected solvable subgroup exist.*

Proof. Projective and affine connected variety can just be a single point. □

Theorem 4.6 (Borel Fixed Point Theorem, BFPT)**.** *Let G be a connected solvable affine algebraic group acting on a proper variety X, then G has a fixed point on X.*

Proof. Induction on $\dim G$. If $\dim G = 0$, then G is trivial, so it is clear. So $\dim G > 0$. Note that G is solvable implies that the derived subgroup $[G, G]$ (i.e. the smallest closed subgroup generated by commutators) is a proper closed normal subgroup. Moreover, it is reduced and connected. Indeed:

$$[G, G] = \overline{\bigcup_{n \geq 1} \operatorname{im}(G^n \times G^n \to G)}$$

Note this union will stop since G is noetherian. Moreover, for each n, the image is a connected constructible set, hence its closure is constructible, connected, with a reduced open piece. Hence we see $[G, G]$ is reduced and connected.

By induction, we know $X^{[G,G]} \neq 0$. This is a closed subvariety of X, hence it is also proper. Moreover $[G, G] \leq G$ normal, G acts on it. So we can replace X by $X^{[G,G]}$, and so $[G, G]$ acts trivially. Now let $x \in X^{[G,G]}$ be such that $G \cdot x$ has minimal dimension. Then $G \cdot x$ is closed, so proper. But

$$G/\mathrm{stab}_G(x) \simeq G \cdot x$$

and $\mathrm{stab}_G(x) \supset [G, G]$, so $\mathrm{stab}_G(x)$ is a normal subgroup in G. But that means $G/\mathrm{stab}_G(x)$ is affine algebraic group. But $G \cdot x$ is proper, affine and connected, hence $G \cdot x = \operatorname{Spec} k$ is a point. Namely, x is fixed by G. □

Proof. (Alternative for BFPT)
Claim: If G is connected reduced affine algebraic group of dimension 1, then $G = \mathbb{G}_a, \mathbb{G}_m$.
Claim: If G reduced solvable connected affine aglebraic group, there exists a normal series $1 < H_1 < H_2 < \cdots < H_r = G$ with $H_i \lhd H_{i+1}$ and each quotient has dimension 1.
Claim: Let $G = \mathbb{G}_a$ or $\mathbb{G}_m$, acting on proper X. Let $x \in X$. If $G \cdot x \neq x$, then we get a map

$$\begin{array}{ccccc} G & \longrightarrow & G \cdot x & \hookrightarrow & X \\ & & \downarrow & \nearrow & \\ & & \mathbb{P}^1 & & \end{array}$$

valuation criterion for properness shows that extends to a map $\varphi : \mathbb{P}^1 \to X$. Show $\varphi(\infty)$ is fixed by G
Claim: Use these claims and inuciton to finish the proof. □

Corollary 4.7 (Lie-Kolchin Theorem)**.** *Let B be a connected solvable closed subgroup of GL_V, then B fixes a flag $(F_1 < F_2 < \cdots < F_{\dim V} = V)$. Hence B is conjugate to a subgroup of upper-triangular matrices.*

Equivalent, if W is an irreducible representation of B, then $\dim W = 1$.

Proof. Apply BFPT to the flag variety of GL_V, which is a projective variety. □

Compare with Lie's theorem in Lie algebra course: Lie's theorem requires $k = \overline{k}$ and $\mathrm{char}(k) = 0$. Here, we have no requirement on the characteristic.

Proof. (of Theorem 4.3, assuming BFPT)

We first prove (2). Choose a Borel B of maximal dimension. Pick a representation (ρ, V) of G with $l \in \mathbb{P}V$, $\mathrm{stab}_G(l) = B$. So $G \cdot l = G/B \subseteq \mathbb{P}V$. Note $\ker \rho$ fixes l, so is a subgroup of B, so is solvable.

Consider $\{l < F_2 < F_3 < \cdots F_{\dim V} = V\} \subset \mathcal{F}l_V = \mathrm{GL}_V/B_V$. This is a closed subvariety, so is projective (and isomorphic to $\mathcal{F}l_{V/l}$). Note B acts on this, so by BFPT we get a fixed point $W_\bullet = (W_1 = l < W_2 < W_3 < \cdots < V)$. Hence $G \cdot W_\bullet \cong G/B$.

Claim: If $F \in \mathcal{F}l_V$ is any flag, then $\dim G \cdot F \geq \dim(G \cdot W_\bullet = G/B)$.

Note this claim immediately shows that $G \cdot W_\bullet$ is closed, hence the second part of the theorem (if B has maximal dimension).

Proof of Claim: Let $\Gamma = \mathrm{stab}_G(F)$. $0 \to \ker \rho \to \Gamma \to \rho(\Gamma) \to 0$. Note $\rho(\Gamma) \subset \mathrm{GL}_V$ fixes a flag, hence is solvable. But $\ker \rho \leq B$ is also solvable, we see that Γ is solvable $\implies$ Γ^0 is solvable. But we choose B of maximal dimension, so $\dim \Gamma \leq \dim B$. But $\dim G \cdot F = \dim G - \dim \Gamma \geq \dim G - \dim B = \dim G/B$. Hence this proves the claim.

We now prove (1). Let B' be any Borel subgroup. G acts on the projective variety G/B, so B' acts on this also. This action has a fixed point by BFPT, say gB. Namely, $b'gB = gB \implies g^{-1}b'g \in B, \forall b' \in B' \implies g^{-1}B'g \subset B$. But conjugate of Borel is Borel, we see $g^{-1}B'g = B$. So all Borel subgroups are conjugate and have the same dimension. □

Lemma 4.8. *Let $P \leq G$ be a closed subgroup. TFAE:*

1. *G/P is projective.*
2. *P contains some Borel subgroup B.*

Proof. If G/P is projective, then B has a fixed point on G/P, say xP. So for any $b \in B$, $bxP = xP \implies x^{-1}bx \in P \implies x^{-1}Bx \subset P$. Conversely, if $P \supset B$ for some Borel B. The map $G \to G/P$ is constant on cosets of B. So by universal property of quotients, it factors through $G/B \twoheadrightarrow G/P$. So we are in the setup that $\varphi : X \to Y$ surjective, X proper. So G/P is proper. But quasiprojective and proper variety is projective. □

Fix $B \subset G$, $\{P | P \supset B, P \text{ reduced and closed}\}$ are called **standard parabolics**. G/P are called **generalized flag varieties**.

Example 4.9. *Let $G = \mathrm{GL}_n, I = \{i_1 < i_2 < \cdots i_r\} \subset \{1, \cdots, n-1\}$. Then $\mathcal{F}l_I = \{F_{i_1} < F_{i_2} < \cdots < F_{i_r} < V | \dim F_k = k\}$ is a generalized flag variety. The stablizer of any generalized flag is a parabolic subgoup. This construction exhausts all parabolics, in the case of GL_n (to see this ,use the correspondence with the parabolic subalgebras of the Lie algebras.)*

4.2 Structure of Solvable Groups

Let G be a connected solvable group.

Lemma 4.10. *Let $x \in [G, G]$, then x is unipotent.*

Proof. Let V be a faithful finite dimensional representation of G. By Lie-Kolchin, G acts by upper triangular matrices. We can thus realize $[G, G]$ concretely, and its elements are clearly unipotent. □

Definition 4.11. A group G is "unipotent" if for all $x \in G$, x is unipotent.

Example 4.12. *The following groups are unipotent:*

1. $U_n \subset \mathrm{GL}_n$, *in particular* $\mathbb{G}_a \subset \mathrm{GL}_2$.
2. *any finite p-group.*

Theorem 4.13 (Kolchin)**.** *If U is unipotent and V is a representation of U. then U fixes a flag in V. In particular, U is nilpotent as a group. Equivalently, every irreducible representation of a unipotent group U is the trivial representation.*

Lemma 4.14 (Wedderburn's Theorem)**.** *Let A/k be a finite dimensional simple algebra, then $A \simeq \mathrm{Mat}_n(D)$, for D a division algebra over k.*

In particular, if $k = \overline{k}$, V finite dimensional vector space over k. Let $A \subset \mathrm{End}_k(V)$ such that V is an irreducible A-module. Then $A = \mathrm{End}_k(V)$.

Proof. (of Kolchin) Let (ρ, W) be an irreducible representation of U. We need to show W is trivial. Let $A = \mathrm{End}_k(W)$ be spanned by $\rho(g), g \in U$.

Define a non-degenerate inner product on A as $\langle x, y\rangle = \mathrm{tr}(xy)$. If $g \in G$, consider $\rho(g) - I \in A$ is nilpotent as g is unipotent. Now consider $\langle \rho(g) - I, \rho(g')\rangle = 0 = \mathrm{tr}(gg') - \mathrm{tr}(g') = 0$, since both gg' and g' are unipotent. So $\rho(g) - I = 0 \implies \rho(g) = I$ for all $g \in G$. This shows W is the trivial representation. □

Remark 4.15. *Compare with Engel's theorem in Lie algebra.*

Definition 4.16. If $G \cong \mathbb{G}_m^a$ for some a, then we call G a **torus**.

Proposition 4.17. *If G is connected solvable group over $k = \overline{k}$.*

1. *$G_u = \{u \in G | u$ unipotent$\}$ is a closed connected normal subgroup of G, containing $[G, G]$. [In particular, $G_u, [G, G]$ are nilpotent, as seen before]*
2. *If G itself is nilpotent as a group, then $G = G_u \times G_s$ a direct product of groups, where $G_s = \{s \in G | s$ is semisimple$\}$ is a subgroup and a torus.*
3. *In general, for G solvable and $T \leq G$ is a maximal torus, there is still an isomorphism*

$$G_u \times T \to G$$

as varieties. Moreover, this is an isomorphism of groups when we give LHS a semidirect product structure. Moreover, all such maximal tori are conjugate.

4. *If $S \leq G$ is any subgroup of of semisimple elements, then S is commutative, and $S \hookrightarrow T$, for some T a maximal torus.*
5. *For such S, $N_G(S) = Z_G(S)$ and is connected.*

Corollary 4.18. *Let G be a connected, reduced, affine algebraic group. Let B be a Borel in G. Choose a maximal torus $T \subset B$. Then we have $B \simeq B_u \rtimes T$. Moreover*

1. *Suppose B', T' is another choice, then there exists $g \in G$ with ${}^g(B,T) = (B',T')$*

2. *The maximal tori of G are all conjugate.*

3. *The maximal connected unipotent subgroups of G are all conjugate.*

Proof. 1. If $g_1 \in G$ such that ${}^{g_1}B = B'$, as all Borels are conjugate. Then ${}^{g_1}T'$ is a maximal torus in B, so equals ${}^{b_1}T$ for some $b_1 \in B$ by (3) above. Set $g = g_1 b_1$.

2. If T is a maximal torus, then T is solvable, so $T \subset B$ in some B. If T' is another such, $T' \subset B'$, then applies (1).

3. Kolchin's theorem tells us a maximal connected unipotent subgroup U is nilpotent as a group, hence is solvable, hence contained in B a Borel. So $U = B_u$ as U is maximal. The result follows by (1).

□

Proposition 4.19. *Let G be connected. $N_G(B) = B$. This means, G/B is the moduli space of all Borels.*

Again let G be an affine algebraic group over k. If H_1, H_2 are both normal closed reduced connected unipotent (resp. solvable) subgroups of G, then so is H_1H_2. Hence there exists a maximal such, called *unipotent radical* (resp. *solvable radical*) of G, denoted as R_uG (resp. RG).

Example 4.20. *1. If G is solvable, then $G_u = R_uG$, $RG = G$.*

2. $R_u(RG) = R_uG$

3. If k'/k is Galois extension, then $R_uG = 1 \iff R_uG_{k'} = 1$.
In fact, $R_uG_k = (R_uG_{k'})^{\mathrm{Gal}(k'/k)}$.

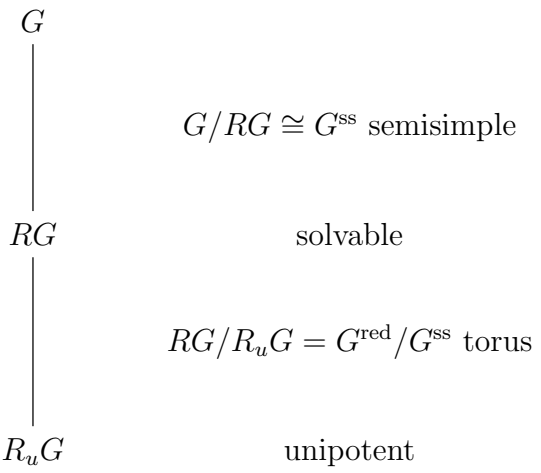
4. This is very false for k'/k inseparable.

Definition 4.21. We call G is *reductive* if $R_uG_{\overline{k}} = 1$. We call G is *semisimple* if $RG_{\overline{k}} = 1$. A reductive group G is *split* if there exists $B \supset T$ defined over k.

Example 4.22. *$G = \mathrm{GL}_n$ is reductive; $\mathrm{SL}_n, \mathrm{PGL}_n$ are semisimple.*
SU_n and $\mathrm{SL}_n(\mathbb{R})$ are both semisimple algebraic groups over $\mathbb{R}$. Note $\mathrm{SL}_n(\mathbb{C})^\sigma = SU_n, \sigma : A \mapsto \overline{A}^{-T}$; $\mathrm{SL}_n(\mathbb{C})^\sigma = SL_n(\mathbb{R}), \sigma : A \mapsto \overline{A}$.
$SL_n(\mathbb{R})$ split, SU_n does not split.

Definition 4.23 (Conrad-Gabber-Prasad)**.** G affine algebraic group. We call G is pseudo-reductive if G is reduced and $R_uG = 1$.

Assume $k = \overline{k}$. Consider $R_uG \lhd RG \lhd G$. Let $G/RG = G^{\mathrm{ss}}$; $G/R_uG = G^{\mathrm{red}}$.



Example 4.24. *Consider k'/k inseparable: $k = \overline{\mathbb{F}}_p((t)), k' = k[y]/(y^p = t) = k((t^{\frac{1}{p}}))$. Let $G = (k')^*$ considered as an algebraic over k ("Weil restriction of G from k' to k").*

We view k as a k' vector space of dimension p. Note $G = k'\backslash\{0\} = \{a \in k^p = k' | N_{k'/k}(a) \neq 0\}$.

We can show that $R_uG = \{1\}$, but $G_{\overline{k}} \xrightarrow{N} \mathbb{G}_{m,\overline{k}}$ has a kernel, a unipotent group of dimension $p-1$. So G is pseudo-reductive, but not reductive. And all pseudo-reductive groups are built from this observation in "evident" ways (CGP).

Let G be affine algebraic group over algebraically closed field k, (B, T) unique up to conjugation.

Theorem 4.25. *Every element of G is in some Borel.*

Corollary 4.26. *Every semisimple element is of G is in some torus; every unipotent element is in some maximal connected unipotent group.*

Proposition 4.27. *If G is reductive, $Z_G(T) = T$. In general, $Z_G(T) = T \cdot V$, for V connected unipotent group, $V \subset R_uG$.*

4.3 Reductive Groups

Assume $k = \overline{k}$.

Definition 4.28. A *character* of an affine algebraic group is a homomorphisms $\chi : G \to \mathbb{G}_m$. Denote $X^*(T)$ as the abelian group of all characters.

Definition 4.29. An affine algebraic group G is *diagonalizable* if for all $g \in G$, g is semisimple and G is commutative.

Lemma 4.30. *TFAE:*

1. *G is diagonalizable*
2. *G is a closed subgroup of some $\mathbb{G}_m^n$*

3. $k[G]$ is spanned by characters.

Proof. "(3) $\implies$ (1)": A character is a map $f : G \to k^*$ group homomorphism. So $\rho_y f = f(y)f \iff$, f is an eigenvector for ρ_y. So if characters span $k[G]$, then ρ_y is semisimple for all $y \in Y$. Moreover, $\rho_y\rho_x = \rho_x\rho_y$ for any $x, y \in G$, so G is commutative by the faithfulness of regular represenation.

"(1) $\implies$ (2)": Let (ρ, V) be a finite dimensional faithful representation of G. Jordan Decomposition tells that $\rho(g)$ is semisimple for all g, and they commute. So there exists $v_1, \cdots, v_n$ a basis of V such that $\rho(g)$ preserving the basis, i.e. $G \subset \mathbb{G}_m^n$

"(2) $\implies$ (3)": Note $k[G] = k[x_1, x_1^{-1}, \cdots, x_n, x_n^{-1}]/I$ for some ideal I. Note each monomial $x_1^{r_1} \cdots x_n^{r_n}$, $r_i \in \mathbb{Z}$, is preserved under $\mathbb{G}_m^N$ (hence G)-action, and they span $k[G]$. □

Exercise. Distinct characters $\chi_j : G \to \mathbb{G}_m$ are linearly independent. In particular, characters span $k[G]$ if and only if they form a basis.

Let $X^*(G) = \mathrm{Hom}(G, \mathbb{G}_m)$ be group of characters of G. If G is diagonalizable, then

$$k[G] = \bigoplus_{\lambda \in X^*(G)} k_\lambda = k[X^*(G)]$$

i.e. $k[G]$ is isomorphic to the group algebra of $X^*(G)$. Since $k[G]$ is finitely generated as an algebra, there exists $\chi_1, \cdots, \chi_m \in X^*(G)$ which generate $k[G]$ as an algebra. Hence $X^*(G)$ is a finitely generated abelian group (generated by $\chi_1, \cdots, \chi_m$).

Lemma 4.31. *TFAE:*

1. G is a torus;

2. G is connected diagonalizable;

3. $k[G] = k[X^*(G)]$, and $X^*(G)$ is a finitely generated free abelian group.

Let $T \subset B$, $\mathfrak{g} = T_1G$, $\mathfrak{b} = \mathrm{Lie}(B)$, $\mathfrak{t} = \mathrm{Lie}(T)$. If V is a finite dimensional representation of T, all elements of T act as semisimple elements, and they commute. So V decomposes into direct sum of 1-dim reps. So

$$V = \bigoplus V_\lambda \quad V_\lambda = \{v \in V | tv = \lambda(t)v, t \in T\}$$

$$X^*(T) = \{\lambda : T \to \mathbb{G}_m\} \cong \mathbb{Z}^N$$

We know $\mathfrak{g} = \bigoplus_{\lambda \in X^*(T)} \mathfrak{g}_\lambda$. If G is *reductive*, then $Z_G(T) = T$ and $\mathfrak{g}_0 = \mathfrak{t}$. So

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\lambda \in X^*(T)} \mathfrak{g}_\lambda$$

If $\mathfrak{g}_\lambda \neq 0$, we say λ is a root of G.

Exercise. 1. G acts on $\mathfrak{g}$ via Lie algebra automorphism: $[\mathrm{Ad}(g)x, \mathrm{Ad}(g)y] = \mathrm{Ad}(g)[x,y]$
In fact, $G^\circ/Z_G = \mathrm{Aut}^0(\mathfrak{g})$.

2. Hence $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$

Example 4.32. *Let* $G = \mathrm{SL}_2, \mathfrak{sl}_2 = \mathfrak{t} + \mathfrak{g}_\alpha + \mathfrak{g}_{-\alpha}$. *Let* $T = \left\{ \begin{pmatrix} t & \\ & t^{-1} \end{pmatrix} : t \in \mathbb{G}_m \right\}$, $\lambda^\vee : \mathbb{G}_m \to T, t \to \begin{pmatrix} t & \\ & t^{-1} \end{pmatrix}$

Define $U_\alpha = \left\{ \begin{pmatrix} 1 & * \\ & 1 \end{pmatrix} \right\}$, $x_\alpha : \mathbb{G}_a \to U_\alpha, t \mapsto \begin{pmatrix} 1 & t \\ & 1 \end{pmatrix}$;

Similarly, define $U_{-\alpha} = \left\{ \begin{pmatrix} 1 & \\ * & 1 \end{pmatrix} \right\}$, *and* $x_{-\alpha} : \mathbb{G}_a \to U_{-\alpha}$.

Notice $\mathrm{Lie}(U_{\pm\alpha}) = \mathfrak{g}_{\pm\alpha}$. *But* $U_\alpha T U_{-\alpha} \cong \mathbb{A}^1 \times \mathbb{G}_m \times \mathbb{A}^1 \hookrightarrow G = \mathrm{SL}_2$ *as a dense open subset (check this is an affine subvariety, complement is defined a single equation). This is called the "big cell".*

Differentiaate $x_{\pm\alpha}$ *and* $\lambda^\vee$ *at* 1 *, we recover* $(E_\alpha = dx_\alpha(1), F_\alpha = dx_{-\alpha}(1), H_\alpha = d\lambda^\vee(1))$ *the standard* $\mathfrak{sl}_2$*-triple, independent of characteristic. In particular, if* $\mathrm{ch}(k) = 2$*, this is not a simple Lie algebra.*

Example 4.33. *Let* $G = \mathrm{PGL}_2$. *Take* $x_{\pm\alpha}$ *as before,* $\mu^\vee : \mathbb{G}_m \to T, t \mapsto \begin{pmatrix} t & \\ & 1 \end{pmatrix}$. *Define* $H' = d\mu^\vee(1) = \begin{pmatrix} 1 & \\ & 0 \end{pmatrix}$. *Then:*

$$[H', E_\alpha] = E_\alpha, [H', F_\alpha] = -F_\alpha, [E_\alpha, F_\alpha] = 2H'$$

If $\mathrm{char}(k) \neq 2$*, then* $\mathfrak{sl}_2 \simeq \mathfrak{pgl}_2$*; they're different otherwise.*

In fact, if G *is semisimple algebraic group and* $\mathrm{char}(k) > 5$*, then* $\mathfrak{g}$ *is a semisimple Lie algebra, but not necessarily otherwise.*

Recall from Lie algebra course:

1. If $\mathfrak{g}$ is a semisimple Lie algebra over $\mathbb{C}$. We know $\mathfrak{sl}_2 \simeq \mathfrak{g}_\alpha + \mathfrak{g}_{-\alpha} + [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}]$
2. $Y = \{\lambda \in \mathfrak{t}^* | \lambda(H_\alpha) \in \mathbb{Z}, \forall \alpha \in \Phi\} \supseteq Y^+ = \{\lambda \in Y | \lambda(H_\alpha) \in \mathbb{Z}_{\geq 0}, \forall \alpha \in \Phi^+\}$. We've seen for any finite dimensional representation V of $\mathfrak{g}$, $V = \bigoplus_{\lambda \in Y} V_\lambda$. Moreover, if V is irreducible, there is a unique highest weight vector λ.

Exercise. For $G = \mathrm{SL}_2$, show $X^*(T) = Y$, where Y is weight lattice for the associated Lie algebra; for $G = \mathrm{PGL}_2$, show $X^*(T) = 2Y \subsetneq Y$.

Definition 4.34 (Coroots)**.** If $\alpha \in \Phi$, define $\alpha^\vee : \mathbb{G}_m \to T, t \mapsto \varphi_\alpha \begin{pmatrix} t & \\ & t^{-1} \end{pmatrix}$. So $\alpha(\alpha^\vee(t)) = t^2$. Let $X_*(T) = \mathrm{Hom}(\mathbb{G}_m, T) = \mathbb{Z}^l$. We have a pairing

$$X^*(T) \times X_*(T) \to \mathbb{Z}$$

defined by

$$\langle \lambda, \mu^\vee \rangle = N, \text{ for } \lambda \circ \mu^\vee(t) = t^N$$

After choosing a set of simple roots, we can thus define the "Cartan Matrix".

Proposition 4.35. *Let G be reductive group, $k = \overline{k}$. Pick $T \subset B \subset G$.*

1. *If $\alpha \in \Phi$, there exists a morphism of algebraic groups $\varphi_\alpha : \mathrm{SL}_2 \to G$ such that $\varphi_\alpha \begin{pmatrix} t & \\ & t^{-1} \end{pmatrix} \subset T, \varphi_\alpha \begin{pmatrix} 1 & t \\ & 1 \end{pmatrix} \subset B$. Define $U_\alpha = \varphi_\alpha \begin{pmatrix} 1 & * \\ & 1 \end{pmatrix}$, $U_{-\alpha} = \varphi_\alpha \begin{pmatrix} 1 & \\ * & 1 \end{pmatrix}$. Moreover $\mathfrak{g}_\alpha = \mathrm{Lie}(U_\alpha)$, $\mathfrak{g}_{-\alpha} = \mathrm{Lie}(U_{-\alpha})$. Note either φ_α is injective or has kernel $\{\pm 1\}$.*

2. *Let $U = R_u B = B_u$. Then $\prod_{\alpha \in \Phi^+} U_\alpha \to U$ as affine varieties. So*

$$\mathrm{Lie}(U) = \eta = \bigoplus_{\alpha \in \Phi^+} \mathfrak{g}_\alpha, B = TU$$

3. *Let $\dot{s_\alpha} = \varphi_\alpha \begin{pmatrix} & 1 \\ -1 & \end{pmatrix}$, so $\dot{s_\alpha}^2 \in T$. Let $W = N(T)/T$. Then W is a finite refelction group, generated by reflections $s_\alpha = \dot{s_\alpha} T/T \in W$. Note W clearly acts on $\mathfrak{t}$, a vector space over k.*

   In fact, W acts on $X^*(T) = \mathbb{Z}^l$ (this is obvious, as W acting on T). But in fact, W acts by reflections, in the sense that $s_\alpha(\beta) = \beta - \langle \alpha^\vee, \beta \rangle \alpha$, defined over $\mathbb{Z}$.

Definition 4.36. The pair $(\Phi, X^*, \Phi^\vee, X_*)$ is called a *root datum*. This refines the Dynkin diagram.

Theorem 4.37. *The root datum determines the pair (G, T, B).*

Exercise. Compute the root datum for classical Lie algebras.

If G is semisimple, there are two canonical root data attached to a Dynkin diagram:

1. There exists a unique G such that $X^*(T) = Y, X_*(T) = \mathbb{Z}\Phi^\vee$. If we're working over $\mathbb{C}$, G is simply connected.

2. There exists a unique G, called the "adjoint group", such that $X^*(T) = \mathbb{Z}\Phi$.

Any G with fixed Dynkin diagram sits between these two:

$$G^{\mathrm{sc}} \twoheadrightarrow G \twoheadrightarrow G^{\mathrm{ad}}, G^{\mathrm{ad}} = G^{\mathrm{sc}}/Z_{G^{\mathrm{sc}}}$$

Remark 4.38. *In fact, $G^{sc}(\mathbb{C})$ is indeed simply-connected topologically. But this is not necessarily true for any field k. Consider $\mathrm{SL}_2(\mathbb{R})$.*

Remark 4.39. *The cardinality of $Z_{G^{sc}}$ is the determinant of the Cartan matrix.*

4.4 Bruhat Decomposition

Let $B \subset T \subset G$, $\mathcal{B} = G/B$. We have

$$B\backslash G/B = G\backslash \mathcal{B} \times \mathcal{B} = N_G(T)/Z_G(T)$$

So this is the Weyl group, if G is reductive. Hence Weyl group parametrizes the orbit, so

$$G = \coprod_{w \in W} B\dot{w}B$$

Example 4.40. *Let $G = \mathrm{GL}_n$, B upper triangular, T diagonal matrices, $W = S_n$. So Bruhat says:*

$$\mathrm{GL}_n = \coprod_{w \in S_n} BwB$$

Proof. Let $g \in \mathrm{GL}_n$, using row operations, making the number of zeroes at the start of each row as large as possible. Necessarily, the number of the zeroes at the start of each row must be distinct. So there exists an element of S_n to make sure the number of zeroes in each row to increase. This means: $wbg \in B$. So indeed $G = \cup_{w \in W} BwB$. We now see the uniqueness. If g is a matrix, we write $\mathrm{supp}(g) = \{(i,j) | x_{ij} \neq 0\}$. If $b_1 \dot{w} b_2 = \dot{w}' \implies b_1 = \dot{w}' b_2^{-1} w^{-1}$. But $\mathrm{supp}(b_1) = \mathrm{supp}(\dot{w}' b_2^{-1} \dot{w}^{-1}) \supset \mathrm{supp}(\dot{w}' \dot{w}^{-1})$. This forces $w = w'$. □

Exercise. Prove Bruhat Decomposition for classical groups.

Proof. (of uniqueness, but better). Let $(F, F') \in \mathcal{F}l^2_{GL_n}$. Let $F = (F_1 \subset F_2 \cdots \subset F_n), F' = (F'_1 \subset F'_2 \cdots \subset F'_n)$. Fix i. Look at the flags $(F'_{i-1} + F_j \cap F'_i)$ as j increases. Now there exists a unique j where the dimension jumps by 1. We then record the (i,j)-th place of a matrix to be 1. This gives a permuation matrix. □

$BwB/B \simeq B/B \cap {}^wB = U/U \cap {}^wU = \prod_{\alpha \in \Phi^+, w^{-1}\alpha \in \Phi^-} U_\alpha \simeq \mathbb{A}^{l(w)}$ is an affine space. In particular, $U^-B/B \simeq \mathbb{A}^{\#\Phi^+} \hookrightarrow G/B$. But $U^-B \hookrightarrow G$ is affine open. So we get the "big cell" U^-B/B affine open, sitting inside $\mathcal{B}$.

The Bruhat decomposition tells us:

$$\mathcal{B} = \coprod_w BwB/B = \coprod_w \mathbb{A}^{l(w)}$$

The biggest piece is what we've constructed above.

So flag varieties are covered by affine spaces: $\bigcup gU^-B$.

Corollary 4.41. *Let $k = \mathbb{C}$*

$$H^{2i}(\mathcal{B}, \mathbb{Z}) = \mathbb{Z}^{\#\{w \in W | l(w) = i\}}, H^{2k+1}(\mathcal{B}, \mathbb{Z}) = 0$$

Example 4.42. *Let $T \subset B \subset G$ be defined over $\mathbb{F}_q$, then $G(\mathbb{F}_q) = \coprod B(\mathbb{F}_q) w B(\mathbb{F}_q)$. So $\#(G/B)(\mathbb{F}_q) = \#G(\mathbb{F}_q)/\#B(\mathbb{F}_q) = \sum_{w \in W} q^{l(w)}$, as B is connected.*

Define $W(t) = \sum_{w \in W} t^{l(w)}$. Show that for $W = S_n$, $W(t) = \prod_{i=1}^n \frac{t^i - 1}{t - 1}$

5 Representation of Semisimple Groups

Let G be semisimple connected algebraic group, $\mathfrak{g}$ its Lie algebra. Choose $T \subset B \subset G$.

5.1 Motivation

Let $k = \mathbb{C}$. If V is an irreducible representation of $\mathfrak{g}$, say $V = L_\lambda$ highest weight representation for $\lambda \in Y = X^*(T^{\mathrm{sc}})$.

Proposition 5.1. *L_λ is in fact an irreducible representation of G^{sc}.*

Remark 5.2. *Actual proof: we will construct L_λ for $\lambda \in X^*(T)^+ = Y^+$ as a representation of G^{sc}, and show as a rep of $\mathfrak{g}$, it is indeed L_λ.*

Proof. If $g \in G$, let $g^*V = V$ be V as a vector space with a twisted $\mathfrak{g}$-action: $x \circ g^*v = g^*(g^{-1}xgv)$. This is an irreducible representation of $\mathfrak{g}$.

But the space of irreducible reps of $\mathfrak{g}$ is a discrete set Y^+. Now G is connected, so this gives a continuous map $G \to Y^+, g \mapsto g^*V$. Hence this must be constant. So $g^*V \cong V$ for all $g \in G$. Schur's lemma implies there is an isomoprhism $\rho_g : g^*V \cong V$, for ρ_g is unique up to $\mathbb{C}^*$. We get a well-defined homomorphism of groups:

$$G \to \mathrm{PGL}(V), g \mapsto \rho_g$$

We then check that this is indeed a homomorphism of algebraic groups. The question is that if we can lift it to a homomorphism $G \to \mathrm{GL}(V)$.

But we can pass to a central extension of G, say $\widetilde{G}$ such that we get $\widetilde{G} \to \mathrm{GL}_V$. However, G^{sc} has no nontrivial central extensions. □

Remark 5.3. *We cannot lift in general. Consider $\mathfrak{g} = \mathfrak{sl}_2 = \mathfrak{pgl}_2$, $V = \mathbb{C}^2$, $G = \mathrm{PGL}_2$.*

Question: Let $V = L_\lambda$, $v = v_\lambda$ is the highest weight vector. What is the stablizer of the highest weight line $kv_\lambda = kv \in \mathbb{P}V$?

Clearly, $\mathfrak{b} \cdot v \subset kv \implies B$ lies in the stabilzer. If $\lambda(H_\alpha) = 0$, then as $E_\alpha v = 0, H_\alpha v = 0 \implies F_\alpha v = 0$.

Let $\Phi_\lambda = \{\alpha \in \Phi | \lambda(H_\alpha) = 0\}$, and $\mathfrak{p}_\lambda = \mathfrak{b} + \sum_{\alpha \in \Phi_\lambda} \mathfrak{g}_\alpha$. We claim the following:

1. $\Phi_\lambda = \mathbb{Z}\{\alpha \in \Pi | \lambda(H_\alpha) = 0\} \cap \Phi$, so corresponds to a subdiagram of the Dynkin diagram.
2. Define $P_\lambda = \langle B, U_\alpha | \alpha \in \Phi_\lambda \rangle$ with Lie algebra $\mathfrak{p}_\lambda$. So P_λ acts on kv via character λ:

$$\begin{array}{ccccc} P_\lambda & \longrightarrow & P_\lambda/[P_\lambda, P_\lambda] & \longrightarrow & \mathbb{C}^\times \\ & & \uparrow\!\!\uparrow & \nearrow_{\lambda} & \\ & & T & & \end{array}$$

3. The stablizer of kv is precisely P_λ. So $G/P_\lambda \cong G \cdot v_\lambda$

Put $Y^{++} = \{\lambda \in X^*(T^{\text{sc}}) | \langle \lambda, \alpha^\vee \rangle > 0, \alpha \in \Phi^+\}$. Then $\lambda \in Y^{++} \iff P_\lambda = B$, $\lambda \in Y^+$. Consider $Y^+_{\mathbb{Q}}$, the non-negative $\mathbb{Q}$-linear span of Y^+. The faces of this simplex correspond to subdiagrams of the Dynkin diagrams, and they in turn correspond to parabolic subgroups of G.

If $\lambda \in Y^+$, $i : G/P_\lambda = G \cdot v_\lambda \hookrightarrow \mathbb{P}L_\lambda$. Note $i^*\mathcal{O}(1)_{v_\lambda} = (kv_\lambda)^*$. This is λ^{-1} as a representation of P_λ.

We expect $i^*\mathcal{O}(1) \coloneqq \mathcal{L}_{-\lambda}$ has an intrinsic definition on G/P_λ in terms of $\lambda : P_\lambda \to \mathbb{G}_m$. Fibre at gP_λ is $g^*(\lambda^{-1})$ of $gP_\lambda g^{-1}$. We get a G-equivariant map

$$L^*_\lambda = \Gamma(\mathbb{P}L_\lambda, \mathcal{O}(1)) \to \Gamma(G/P_\lambda, i^*\mathcal{O}(1) = \mathcal{L}_{-\lambda})$$

We expect this map to be an isomorphism (note this is obviously injective, as L^*_λ irreducible). So $\mathcal{L}_{-\lambda}$ is a very ample line bundle on G/P_λ.

5.2 Line Bundles and Representations

Let $k = \overline{k}$, G semisimple, $T \subset B \subset G$. We have the following agenda:

1. For all $\lambda \in X^*(T)$, construct a line bundle $\mathcal{L}_\lambda$ on G/B.
2. Show $\mathcal{L}_\lambda$ is very ample on G/P_λ, if $\lambda \in X^*(T)^+$
3. Show $\Gamma(G/B, \mathcal{L}_\lambda) = \Gamma(G/P_\lambda, \mathcal{L}_\lambda)$ is an indecomposable reps, if $\lambda \in Y^+$. Moreover, its dual is a highest weight module with highest weight λ. In particular, this is irreducible for $k = \mathbb{C}$.
4. Describe higher cohomologies.

We have an equivalence of categories

$$B - \text{modules} \simeq (G - \text{equivariant quaiscoherent sheaves on } \mathcal{B} = G/B) \coloneqq \text{QCoh}_G(G/B)$$

with the following properties $M \mapsto G \times^B M \coloneqq \mathcal{M}$ such that

1. fibre of $\mathcal{M}$ at B is M as a B-module.
2. $\Gamma(G/B, \mathcal{M}) = \{f : G \to M | f(gb) = b^{-1}f(g), b \in B, g \in G\} \subset k[G] \otimes M = \ker\{k[G] \otimes M \to k[B \times G] \otimes M\} \in \text{Rep } G$. More generally, let $U \subset G/B$ affine open, let $\widetilde{U}$ be its inverse image in G, then $\Gamma(U, \mathcal{M}) = \{f : \widetilde{U} \to M | f(gb) = b^{-1}f(g)\} = \ker(k[\widetilde{U}] \otimes M \to k[B] \otimes k[\widetilde{U}] \otimes M) \subset k[\widetilde{U}] \otimes M$.

Note $\Gamma(\mathcal{B}, \mathcal{M})$ is finite dimensional, as, G/B is projective. If $\lambda \in X^*(T)$, k_λ be one dimensional representation of B. Pick $v_\lambda \in k_\lambda$. We get a line bundle $\mathcal{L}_\lambda$ on $\mathcal{B}$, hence $H^0(\lambda) = \Gamma(G/B, \mathcal{L}_\lambda)$ a finite dimensional representation of G.

Remark 5.4. *If V is any reps of algebraic group G, then $V = \bigoplus V_\lambda$ weight space decomposition. For $w \in N(T)/T$, and pick a lift $\dot{w}$. Then $V_\lambda \to V_{w\lambda}, v \mapsto \dot{w}v$ is an isomorphism. So* $\text{ch}(V) \in k[[X^*(T)]]$ *is Weyl group invariant.*

Lemma 5.5. *Let $\lambda \in X^*(T)$ such that $\Gamma(\mathcal{B}, \mathcal{L}_\lambda) = H^0(\lambda) \neq 0$. Then*

1. *$H^0(\lambda)^{U^-} = H^0(\lambda)_\lambda$ is one dimensional.*
2. *$H^0(\lambda)_\mu \neq 0 \implies w_0\lambda \leq \mu \leq \lambda$, where w_0 is the unique element of longest element, and $\lambda \geq \mu$ means $\lambda - \mu = \sum_{\alpha_i \in \Pi} k_i(-\alpha_i)$, $k_i \geq 0$.*
3. *$H^0(\lambda)$ is indecomposable, with a simple socle (maximal semisimple submodule), which we will denote $L(\lambda)$.*

Proof. 1. We have the big cell $U^-B \subset G$, $U^- \hookrightarrow \mathcal{B}$ inclusion of affine open. So have a restriction map

$$\Gamma(\mathcal{B}, \mathcal{L}_\lambda) \to \Gamma(U^-, \mathcal{L}_\lambda) = k[U^-] \otimes k_\lambda$$

If $f \in \Gamma(\mathcal{B}, \mathcal{L}_\lambda)$, $f(u^-tu) = \lambda(t^{-1})f(u^-)$. So the above map is equivariant for $U^-T = B^-$-action. And its an injection, as U^- is dense open.

[Compare with Lie algebra, $M_\lambda \twoheadrightarrow L_\lambda$, for M_λ Verma module]

So $\Gamma(\mathcal{B}, \mathcal{L}_\lambda)^{U^-} \subset (k[U^-] \otimes k_\lambda)^{U^-} = k_\lambda$. But for any finite dimensional reps of G, $V^{U^-} \neq 0$ by Kochin's theorem.

2. If μ is a weight of $H^0(\lambda)$, μ is a weight of $k[U^-] \otimes k_\lambda \implies \mu = \lambda - \gamma, \gamma$ is a weight of $k[U^-] \implies \gamma = \sum_{\alpha_i \in \Pi} k_i(-\alpha_i), k_i \geq 0$. Now if μ is a weight of $H^0(\lambda)$, then $w_0\mu$ is a weight of $H^0(\lambda)$, so $w_0\mu \leq \lambda$ [Exercise: $w_0\mu \leq \lambda \iff w_0\lambda \leq \mu$].

3. If $L \subset H^0(\lambda)$ is a simple submodule. So $0 \neq L^{U^-} \subset H^0(\lambda)^{U^-}$. So $L^{U^-} = H^0(\lambda)_\lambda$. But then L is the smallest G-submodule containing $H^0(\lambda)_\lambda$. So L is unique, in particular, $H^0(\lambda)$ is indecomposable.

□

Corollary 5.6. *$H^0(\lambda)$ has a composition series with factors $L(\lambda)$ and simple modules with weights in $\{\mu | w_0\lambda < \mu < \lambda\}$.*

Corollary 5.7. *If V is a simple G-module, $V \simeq L(\lambda)$ for a unique $\lambda \in X^*(T)$ with $H^0(\lambda) \neq 0$.*

Proof. Choose λ such that $\mathrm{Hom}_B(V, k_\lambda) \neq 0$. But now we have "Frobenius Reciprocity" (exercise):

$$\mathrm{Hom}_G(V, \Gamma(\mathcal{B}, \mathcal{L}_\lambda)) = \mathrm{Hom}_B(V, k_\lambda)$$

Hence we have a nonzero map $V \to H^0(\lambda)$. But V is irreducible, so is an injection. Hence $V = L(\lambda)$ by previous result. λ is unique since $V^{U^-} = k_\lambda$. □

Exercise. Show $L(\lambda)^* = L(-w_0\lambda)$

Let $X^+ = \{\lambda \in X^*(T) | \langle \alpha^\vee, \lambda \rangle \geq 0, \forall \alpha \in \Phi^-\}$ be the set of dominant weights (this is an unstandard choice).

Proposition 5.8. $\lambda \in X^+ \iff H^0(\lambda) \neq 0$

Proof. If $H^0(\lambda) \neq 0 \implies H^0(\lambda)_\lambda \neq 0 \implies H^0(\lambda)_{s_\alpha \lambda} \neq 0, \forall \alpha \in \Phi \implies s_\alpha \lambda \leq \lambda \implies \lambda \in X^+$. Conversely, Bruhat decomposition implies

$$\mathcal{B} = \coprod_{w \in W} (U^- wB)/B = (U^- B/B) \coprod (\coprod_{\alpha \text{ simple}} U^- s_\alpha B/B) \coprod (\coprod_{l(w)>1} U^- wB/B)$$

So to find a section $f \in \Gamma(G/B, \mathcal{L}_\lambda)$, i.e. finds $f \in k[U^-] \otimes k_\lambda$ which extends.

We want to use "Hartog's Principle": if X is normal, then $\Gamma(U, \mathcal{L}) \simeq \Gamma(X, \mathcal{L})$, if $\operatorname{codim}(X - U) \geq 2$.

First of all prove the statement of $G = \mathrm{SL}_2$. Indeed $\mathcal{B} = (U^- B/B) \coprod (U^- sB/B) = \mathbb{A}^1 \coprod \{*\}$. Now $v_n \in k[\mathbb{A}^1] \otimes k_n$ extend if and only if $n \leq 0$.

In general, define $U^-_{\neq \alpha} = \prod_{\beta \in \Phi^+, \beta \neq \alpha} U_{-\beta}$ (this is a subgroup if $\alpha \in \Pi$), so $U^- = U^-_{\neq \alpha} \times U_{-\alpha}$. We have an embedding

$$U^-_{\neq \alpha} \times \mathbb{P}^1 = U^-_{\neq \alpha} \times ((U_{-\alpha} B/B) \coprod (U_{-\alpha} s_\alpha B/B)) = (U^- B/B) \coprod (U^- s_\alpha B/B) \to \mathcal{B}$$

So $v_\lambda \in k[U^-] \otimes k_\lambda$ extends to $U^-_{\neq \alpha} \times \mathbb{P}^1$ if and only if $\langle \lambda, \alpha^\vee \rangle \geq 0$ for $\alpha \in -\Delta$ by the above SL_2-computation. □

Corollary 5.9. $\{\text{ch } H^0(\lambda) \mid \lambda \in X^+\}$ *form a basis of* $(\mathbb{Z}[X^*(T)])^W$.

So do $\{\text{ch } L(\lambda), \lambda \in X^+\}$.

Proof. Note $\{m_\lambda = \sum_{\mu \in W \cdot \lambda} e^\lambda \mid \lambda \in X^+\}$ for a basis of $(\mathbb{Z})[X^*(T)]^W$ as every $\lambda \in X^*(T)$ is conjugate under W to a unique $\lambda \in X^+$. But ch $L_\lambda = m_\lambda + \sum_{\mu < \lambda} a_{\lambda\mu} \mu$, so we can do a change of basis to put things on upper triangular. □

Recall if $\mathcal{L}$ is ample line bundle on X, then there exits an embedding:

$$X \hookrightarrow \mathbb{P}V$$

such that $i^* \mathcal{O}(1) = \mathcal{L}^m$. If $m = 1$, we call it very ample.

Proposition 5.10. *Let* $X^{++} = \{\lambda \in X^*(T) \mid \langle \alpha^\vee, \lambda \rangle > 0, \forall \alpha \in -\Phi^+\}$. *Then* $\lambda \in X^{++} \iff$ $\mathcal{L}_\lambda$ *is an ample line bundle.*

Proof. " $\Longleftarrow$ ": Let α be any positive root. Consider $\varphi_\alpha : \mathrm{SL}_2 \to G$, inducing $\overline{\varphi_\alpha} : \mathbb{P}^1 \to \mathcal{B}$. We check that $\overline{\varphi_\alpha}^* \mathcal{L}_\lambda = \mathcal{O}(-\langle \lambda, \alpha^\vee \rangle)$. This must be an ample line bundle on $\mathbb{P}^1$, so $\lambda \in XX^{++}$.

" $\Longrightarrow$ ": Fix $\lambda \in X^{++}$. Let $G/B \hookrightarrow \mathbb{P}V$ for some V, with $B = \mathrm{stab}_G(l)$, where $l = kv \in \mathbb{P}V$. Note U acts trivially on l, so B acts via $B \twoheadrightarrow T \xrightarrow{\mu} \mathbb{G}_m$, for some character μ. Given such μ, there exists some $m > 0$ such that $\mu + m\lambda \in X^+$. Now consider $W = L(\mu + m\lambda)$, so $W^U = kw$ has weight $-\mu - m\lambda$. We look at the G-action on $k \cdot (v \otimes w) \in V \otimes W'$. Note B acts as $m\lambda$. So indeed $\mathrm{stab}_G(k \cdot (v \otimes w)) = B$. Therefore we get:

$$i : G/B \simeq G \cdot (v \otimes w) \hookrightarrow \mathbb{P}(V \otimes W)$$

with $\mathcal{L}_\lambda^{\otimes m} = \mathcal{L}(m\lambda) = i^* \mathcal{O}_{\mathbb{P}(V \otimes W)}(1)$. So $\mathcal{L}_\lambda$ is ample. □

Remark 5.11. *In fact, one can show that $\mathcal{L}_\lambda$ is very ample.*

Exercise. 1. If $\lambda \in X^+$, then $\mathcal{L}_\lambda$ is ample on G/P_λ

2. If $\lambda \in X^+$, then $H^i(G/P_\lambda, \mathcal{L}_\lambda) = H^i(G/B, \mathcal{L}_\lambda)$.

Proposition 5.12. *If G is simply connected, then $X^*(T) \simeq \mathrm{Pic}\ \mathcal{B}$ via $\lambda \mapsto \mathcal{L}_\lambda$.*

Proof. If $\mathcal{L} \in \mathrm{Pic}\ \mathcal{B}$, set $\overline{\varphi_\alpha} : \mathbb{P}^1 \to \mathcal{B}$ as above, for α a simple root. Suppose $\overline{\varphi_\alpha}^*\mathcal{L} = \mathcal{O}(n_\alpha)$. Write $\lambda = \sum n_\alpha w_\alpha$, for $w_\alpha \in X^*(T)$ the fundamental weights. Set $\mathcal{M} = \mathcal{L} \otimes \mathcal{L}_\lambda^{-1}$, we have $\overline{\varphi_\alpha}^*\mathcal{M} = \mathcal{O}$ for all simple root α.

But $H^1(\mathcal{B}, \mathbb{Z}) = 0$, $H^2(\mathcal{B}, \mathbb{Z}) \cong \mathbb{Z}^{\#\Delta}$ from Bruhat decomposition. So the first Chern Classes $c_1(\mathcal{L}_{w_\alpha})$ form a basis of $H^2(\mathcal{B}, \mathbb{Q})$, and $c_1(\mathcal{M}) = 0 \iff \mathcal{M} = \mathcal{O}$. □

5.3 Weyl Character Formula

Definition 5.13. Let $\lambda \in X = X^*(T)$, the *Weyl Character* associated with λ is:

$$\chi_\lambda = \sum_{w \in W} w\left(\frac{e^\lambda}{\prod_{\alpha>0}(1 - e^{-\alpha})}\right) = \frac{\sum_{w \in W}(-1)^{l(w)} e^{w(\lambda+\rho)-\rho}}{\prod_{\alpha>0}(1 - e^{-\alpha})}$$

, where $\rho = \frac{1}{2}\sum_{\alpha>0} \alpha$.

Remark 5.14. *Note that $\mathcal{L}(-2\rho) = \det T^*\mathcal{B}$. Indeed, $T_B(G/B) = \mathfrak{g}/\mathfrak{b} \implies T_B^* G/B = (\mathfrak{g}/\mathfrak{b})^*$. So $\det T_B^*(G/B) = \bigwedge^{\mathrm{top}}(\mathfrak{g}/\mathfrak{b})^*$ has weights $\sum_{\alpha \in \mathrm{wt}(\mathfrak{g}/\mathfrak{b})} -\alpha = -2\rho$.*

Definition 5.15.

$$\mathrm{ch}H^*(\mathcal{B}, \mathcal{L}_\lambda) = \sum (-1)^i \mathrm{ch}H^i(\mathcal{B}, \mathcal{L}_\lambda) \in \mathbb{Z}[X]^W$$

Proposition 5.16.

$$\mathrm{ch}H^*(\mathcal{B}, \mathcal{L}_\lambda) = \chi_\lambda$$

if $\lambda \in X$

Remark 5.17. *This has a nice algebraic-geometric proof. Let T be a torus acting on a smooth projective vareity X, with isolated fixed points. Suppose $\mathcal{V}$ is a T-equivariant vector bundle on X, then we have "localization to the fixed points":*

$$\mathrm{ch}H^*(X, \mathcal{V}) = \sum_{x \in X^T} \mathrm{ch}\left(\frac{\mathcal{V}|_x}{\prod_{\alpha \in \mathrm{wt}(T_x X)}(1 - e^{-\alpha})}\right)$$

In our case, consider $X = \mathcal{B}$ with T-action. The fixed points are of form wB for $w \in W$.

Proof. Consider the W-action on X: $w * \lambda = w(\lambda + \rho) - \rho$. Note:

$$\chi_\lambda \cdot \prod_{\alpha>0}(1 - e^{-\alpha}) = \sum_{w \in W}(-1)^{l(w)} e^{w*\rho}$$

is obviously $*$-anti-invariant. So χ_λ is $*$-anti-invariant. Moreover χ_λ is W-invariant, and $\chi_\lambda = e^\lambda + \sum_{\mu<\lambda} m_{\mu\lambda} e^\mu$ for some $m_{\mu\lambda} \in \mathbb{Z}_{\geq 0}$. By a combinatorial argument we see that these properties *characterize* χ_λ. It remains to show that $\mathrm{ch}(H^*(\mathcal{B}, \mathcal{L}_\lambda))$ also has these properties
. □

Proposition 5.18 (Kempf Vanishing Theorem)**.** *If $\lambda \in X^+$, $H^i(\mathcal{B}, \mathcal{L}_\lambda) = 0$ for all $i > 0$.*

Remark 5.19. *For characteristic 0 case, this is an immediate consequence of Kodaira vanishing theorem.*

Note if $\lambda \in X^+$, then $\mathrm{ch}(H^0(\lambda)) = \chi_\lambda$. So if $\mathrm{char}(k) = 0$, then $H^0(\lambda)$ indecomposable $\implies H^0(\lambda) = L(\lambda)$ as representations are semisimple (which we did not prove). This gives Weyl character formula for Lie algebras. If $\mathrm{char}(k) = p > 0$, we don't know $\mathrm{ch}(L(\lambda))$, but still know $\mathrm{ch}(H^0(\lambda))$.

5.4 Borel-Weil-Bott

Consider the W-action on X: $w * \lambda = w(\lambda + \rho) - \rho$. One easily sees this action is anti-invariant, i.e. for any simple reflection s we have $\chi_{s*\lambda} = -\chi_\lambda$.

Proposition 5.20. *Let $s = s_\alpha$ be a simple reflection.*

1. $\mathrm{ch}(H^*(\mathcal{B}, \mathcal{L}_\lambda)) = -\mathrm{ch}(H^*(\mathcal{B}, \mathcal{L}_{s*\lambda}))$
2. *If $\mathrm{char}(k) = 0$ and $\langle \alpha^\vee, \lambda \rangle \geq -1$, then:*

$$H^i(\mathcal{B}, \mathcal{L}_\lambda) = H^{i+1}(\mathcal{B}, \mathcal{L}_{s*\lambda})$$

Moreover, if $\langle \alpha^\vee, \lambda \rangle = -1$, then $H^i(\mathcal{B}, \mathcal{L}_\lambda) = 0$ for all i .

Corollary 5.21 (Borel-Weil-Bott)**.** *Let $\mathrm{char}(k) = 0$.*

1. *Let $\lambda \in X^+$, then:*

$$H^i(\mathcal{B}, \mathcal{L}_{w*\lambda}) = \begin{cases} H^0(\mathcal{B}, \mathcal{L}_\lambda) & \text{if } i = l(w); \\ 0, & \text{otherwise.} \end{cases}$$

2. *If $\lambda \in X \backslash W * X^+$, then there exits $\alpha \in \Phi$ such that $\langle \lambda + \rho, \alpha^\vee \rangle = 0$, and $H^*(\mathcal{B}, \mathcal{L}_\lambda) = 0$.*

Proof. Claim: If $i < l(w)$, then $H^i(\mathcal{B}, \mathcal{L}_{w*\lambda}) = 0$.
Proof of Claim: We prove by induction on $l(w)$. The case for $l(w) = 0$ is clear.

Now assume $l(w) > 0$. Pick $s = s_\alpha$ simple such that $s_\alpha w < l(w)$, then $-w^{-1}\alpha \in \Phi^+$. But then

$$\langle s_\alpha(w * \lambda), \alpha^\vee \rangle = \langle w(\lambda + \rho), -\alpha^\vee \rangle - 1 = \left\langle \lambda + \rho, -w^{-1}\alpha^\vee \right\rangle - 1 \geq 0$$

So by the above proposition, we know:

$$H^i(\mathcal{B}, \mathcal{L}_{w*\lambda}) = H^{i-1}(\mathcal{B}, \mathcal{L}_{s_\alpha(w*\lambda)}) = 0$$

by induction hypothesis. This proves the claim.
Claim: If $i > l(w)$, $H^i(\mathcal{B}, \mathcal{L}_{w*\lambda}) = 0$
Proof of Claim: We perform decreasing induction on $l(w)$, starting from $w = w_0$. Note $H^i(\mathcal{B}, \mathcal{L}_\lambda) = 0$ for all $i > l(w_0) = \dim \mathcal{B}$ by Serre's vanishing theorem.

With these two claims, the above corollary follows immediately. □